Chapter 9

Classical and emerging approximations for the screening of antimicrobial peptide libraries

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1 Introduction

1.1 Importance of screening peptide libraries

Peptides stand as versatile components within the pharmaceutical realm, demonstrating diverse utilities from active pharmaceutical ingredients to aiding drug discovery and delivery (Rhym et al., 2023; Schissel et al., 2021). Their distinctive traits, including tissue penetration, high-affinity interactions with receptors, and antimicrobial properties, render them invaluable in these applications (Bozovičar & Bratkovič, 2019). Antimicrobial peptides (AMPs) serve as a promising class of therapeutic agents, effective against a wide range of pathogens, including bacteria, fungi, and viruses (Rodríguez-Soto et al., 2021). Thus, peptide libraries—collections of concise amino acid sequences—emerge as indispensable tools in drug discovery, protein engineering, and fundamental research (Lerksuthirat et al., 2023). These libraries offer a plethora of potential ligands capable of interacting with proteins in highly specific ways (Righetti & Boschetti, 2013). By generating and screening extensive peptide collections, researchers pinpoint molecules with significant biological activities or binding affinities (Madden, 2021; Sohrabi et al., 2020). The screening process typically entails exposing the peptide library to a target molecule, such as a receptor or enzyme, and subsequently identifying the peptides that interact with the target (Bozovičar & Bratkovič, 2019). These applications underscore the pivotal role peptides and peptide libraries play in both the pharmaceutical industry and scientific exploration (Fig. 9.1).

The screening of peptide libraries enables the rapid identification of novel peptides possessing desired properties. A key advantage lies in swiftly identifying bioactive peptides, as demonstrated across various studies (Conibear et al., 2020; Kaur et al., 2015; Liu et al., 2017). Moreover, these libraries contribute significantly to discovering novel targets, crucial in developing new therapies and drugs (Lerksuthirat et al., 2023; Li, Pu, et al., 2023). Furthermore, optimizing peptide structure through library screening enables the identification of peptides with enhanced bioactivity or selectivity (Hashemi et al., 2021; Kaguchi et al., 2023; Schatz, 1993). Additionally, screening peptide libraries serve in validating binding assays, showcasing their capacity to offer prompt and reliable results (Ivarsson, 2022). Overall, the utilization of screening peptide libraries spans a wide range of applications, significantly contributing to the development of novel therapeutics and diagnostics. However, the intricate and challenging nature of screening peptide libraries necessitates effective methods for identifying and characterizing the most promising candidates.

In this chapter, we embark on a comprehensive exposition delineating classical and emerging approximations for screening peptide libraries. Our focus encompasses elucidating approaches to monitor the screening process in real time. This chapter endeavors to encompass established methodologies grounded in extensive and burgeoning miniaturized systems for screening peptide libraries. Additionally, we aim to illuminate the most critical technical challenges while presenting future perspectives in this domain.

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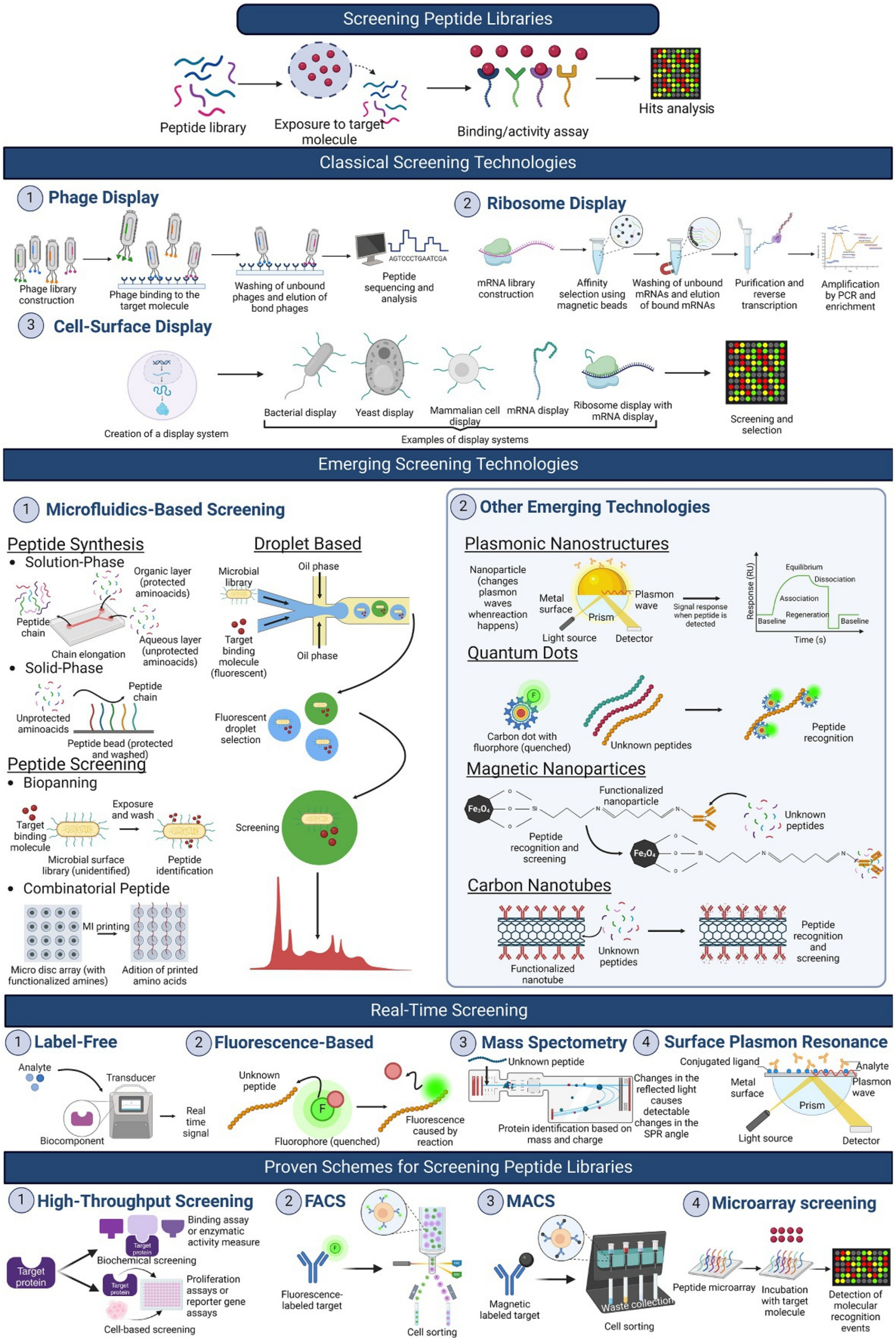

FIG. 9.1 Graphical abstract of classical and emerging approximations for the screening of peptide libraries. An encompassing schematic representation delineates the array of peptide library screening methodologies, encapsulating classical techniques, including phage display, ribosome display, and cell-surface display, as well as emergent strategies such as microfluidics-based screening and nanotechnology applications. It further portrays real-time screening modalities, embracing label-free, fluorescence-based, mass spectrometry, and surface plasmon resonance techniques, concluding with a compendium of established screening schemas like high-throughput screening, fluorescence-activated cell sorting, magnetic-activated cell sorting, and microarray screening, thus providing a synthesized panorama of the stratagems employed in the identification of bioactive peptides. A visual summary showcasing various peptide library screening methods, from traditional displays and microfluidic systems to real-time detection techniques and established screening protocols, highlighting the process of identifying active peptides.

1.2 Overview of classical and emerging approximations for screening peptide libraries

Peptide libraries are a powerful tool for identifying peptides with specific biological activities or binding affinities. Classical methods for screening peptide libraries include phage display, ribosome display, cell surface display (including bacterial display, yeast display, and mammalian display), and cDNA/mRNA display (including phage, yeast, and ribosome display) (Sohrabi et al., 2020; Ullman et al., 2011). Phage display involves the presentation of peptides on the surface of bacteriophages, which are then screened for binding to specific targets (Pande et al., 2010). Ribosome display uses in vitro transcription and translation to display peptides on ribosomes, which are then screened for activity (Li, Kang, et al., 2019). Cell surface display involves the expression of peptides on the surface of cells, which are then screened for binding or activity (Lee et al., 2003). cDNA/mRNA display involves the attachment of peptides to the 3′ end of a DNA or RNA molecule, which is then translated in vitro to produce a library of peptides (Ueno and Nemoto, 2012). Synthetic peptide libraries can also be screened using methods such as CAD and CIS display or SNAP-tag display (Ullman et al., 2011). While classical methods have been successful in identifying important peptide-based therapeutics, they are often labor-intensive, time-consuming, and require large amounts of starting material (Audie & Boyd, 2010; Haney & Hancock, 2013).

More recently, microfluidic systems have emerged as promising alternatives to classical methods due to their high throughput, low cost, and small sample volumes (Hung et al., 2014). Microfluidic chips for peptide library synthesis and screening allow for precise control over reaction conditions, while droplet-based microfluidics enable the encapsulation of individual peptides in picoliter-sized droplets for screening (Dressler et al., 2014). Millireactors, which allow for continuous flow reactions, have also been used for peptide library screening. Nanotechnology-based platforms, such as nanopore sensors and quantum dot (QD)-based sensors, have also shown great potential for screening peptide libraries with high sensitivity and specificity (George et al., 2022). Hybrid approaches have also been developed to enhance the capabilities of classical and emerging methods. For example, microarrays-nanotechnology with phage display combines the high-throughput capabilities of microarrays with the sensitivity of nanotechnology-based sensors and the versatility of phage display (Goracci et al., 2020). Machine learning algorithms have also been applied to improve the efficiency and accuracy of peptide library screening (Plisson et al., 2020).

In summary, classical and emerging approximations for screening peptide libraries have greatly expanded the scope and capabilities of peptide-based research. The development of new methods and the integration of existing methods have resulted in more efficient, accurate, and high-throughput screening (HTS) of peptide libraries, leading to the identification of novel peptide-based therapeutics and diagnostics.

2 Classical screening technologies for peptide libraries

2.1 Phage display

George P. Smith first described the concept of phage display in a report in 1985 (Smith, 1985), and later, the concept of phage-displayed peptide libraries in an article in 1988 (Parmley & Smith, 1988). As a result of the flurry of activity that inspired his concept, the first phage display libraries were published in 1990 (Cwirla et al., 1990; Devlin et al., 1990; Scott & Smith, 1990). Phage display is a very effective and extensively used approach for identifying and isolating peptides with specific binding characteristics from large peptide libraries (Newton & Deutscher, 2008; Paschke, 2006; Sidhu, 2000; Wu et al., 2016). This technology allows the expression of exogenous (poly)peptides on the surface of phage particles. The display of peptides on the surface of bacteriophages, which are viruses that infect bacteria, is how phage display works. The foreign DNA fragment is integrated into the filamentous phage's genome, and the encoded foreign peptide is displayed as a fusion to one of the phage's coat proteins on the surface, usually pIII, pVI, or pVIII (Hoogenboom et al., 1998; Kramer, 2003; Mullen et al., 2006; Sandman et al., 2000). The physical attachment of the displayed peptide to the viral particle, together with the notion of combinatorial peptide libraries, has established phage display as a tool for developing novel ligands with desired properties (Pande et al., 2010). Traditional phage display screening approaches often require many steps, including library building, library screening (i.e., phage amplification, target immobilization, phage binding, washing, and elution), and further analysis (Azzazy and Edward Highsmith, 2002; Hoogenboom et al., 1998; Marintcheva, 2018; Mimmi et al., 2019). The following is a high-level summary of the steps involved in phage display screening.

- *Library construction*: By far, the most popular phage that has been used for display is the filamentous bacteriophage. Inserting random or planned peptide sequences into the coat protein gene of a bacteriophage, generally M13 (Naik et al., 2002; Noren & Noren, 2001; Rodi et al., 2002; Sidhu, 2001; Sugimura et al., 2006) or T7 phage (Caberoy et al., 2009;

Danner & Belasco, 2001; Piggott & Karuso, 2016) but also T4, and Lambda (Marintcheva, 2018), produces a varied library of peptides. As a result, there is a large population of phage particles with diverse peptides on their surface.

- *Phage amplification*: Filamentous phages are propagated in pilus-positive bacteria (e.g., *Escherichia coli*) that are not lysed by the phage but secrete multiple copies of phage displaying a particular insert. For ligand display, many filamentous phage coat proteins have been utilized. The pIII phage protein, which is important in bacterial infection and is present in three to five copies per phage particle, is by far the most often utilized (Hoogenboom et al., 1998).
- *Target immobilization*: The target of interest, such as a protein or a receptor, is immobilized onto a solid surface, such as a microplate or a bead. This allows the target to be captured and retained during the screening process.
- *Phage binding*: The phage library is added to the immobilized target, and the phage particles that display peptides with binding affinity to the target will bind specifically to the target.
- *Washing*: Unbound phage particles are removed by washing the solid surface to reduce background noise and increase the specificity of the screening.
- *Elution*: The bound phage particles are then eluted, typically by changing the conditions, such as altering the pH or using competitive elution buffers, to release the phage particles from the target while retaining the phage particles that specifically bind to the target.
- *Amplification and enrichment*: The eluted phage particles are then amplified usually in *E. coli* to generate a new generation of phage particles, and the process of binding, washing, and elution can be repeated through multiple rounds to enrich for phage particles that bind specifically to the target.
- *Peptide sequencing and analysis*: After several rounds of screening, the enriched phage particles are sequenced to identify the peptide sequences that are responsible for the specific binding to the target. These peptide sequences can then be synthesized and further characterized for their binding properties.

Peptide screening has been transformed by phage display for peptide libraries, which allows for the rapid screening of large peptide libraries and the identification of peptides with specific binding characteristics (Krumpe & Mori, 2006; Yang et al., 2022). It has been widely used in a variety of fields, including drug discovery (Mangini et al., 2017; Mimmi et al., 2019; Robson & Ghatage, 2011), protein engineering (Dwyer et al., 2000; Hoess et al., 2001; O'Neil and Hoess, 1995; Sidhu & Koide, 2007), and diagnostics (Anand et al., 2021; Deutscher, 2010; Schirrmann et al., 2011), and has played an important role in the discovery of critical peptides with important biological roles, such as ligands (Katz, 1995, 1997; Nixon, 2002; Shen et al., 2013; Turnbough, 2003), mimotopes (Cortese, 2001; Lin & Wu, 2004; Luzar et al., 2016; Petrenko et al., 2022; Ru et al., 2010; Zhou et al., 2021), and protein-protein interaction motifs (Davey et al., 2017; Seo & Kim, 2018; Sundell & Ivarsson, 2014). Furthermore, advances in phage display technology have been made, including the use of different types of phages, peptide modifications, and the creation of novel phages (Huang et al., 2022). Peptide libraries have facilitated the discovery of potential lead compounds for developing novel peptidomimetic drugs by allowing the characterization of peptide binding specificity in important proteins such as antibodies involved in inflammatory reactions (Alfaleh et al., 2020; Frenzel et al., 2016; Ladner et al., 2004) or integrins that mediate cellular adherence (Nowakowski et al., 2004; Ramaraju et al., 2017; Romanov et al., 2001). Phage libraries are also widely used for displaying not only small peptides but also larger protein domains such as single-chain antibodies (Boel et al., 2000; Eisenhardt et al., 2007; Mao et al., 1999; Rondot et al., 2001; Tang et al., 1996). However, as with any screening approach, careful experimental design and optimization are required to produce accurate and relevant results.

2.2 Ribosome display

Ribosome display is an excellent technology for selecting functional proteins and peptides from large libraries. It is based on the production of a ternary complex of messenger RNA (mRNA)-ribosome-nascent polypeptide in a cell-free protein synthesis system (Hanes et al., 2000; Hanes & Plückthun, 1997). In brief, a randomized peptide library encoded by an mRNA template is translated in vitro using ribosomes, which are cellular structures responsible for protein synthesis. The peptides are synthesized by the ribosomes, and the nascent peptides remain attached to the ribosomes via a peptidyl-tRNA linkage (Marr et al., 2011). This allows the peptides to be displayed on the ribosome surface, where they can be screened and selected (Gersuk et al., 1997; Mattheakis et al., 1994, 1996). Ribosome display methods, which are entirely in vitro, have some advantages over selection technologies that involve a step in cells, such as phage display. The greatest advantage is that the library's diversity is not limited by cell transformation efficiency (Ohashi et al., 2012).

The first experimental demonstration of ribosome display technology was the selection of short peptides from a library using an in vitro translation system based on *E. coli* S30 extract match (Mattheakis et al., 1994, 1996). However, because *E. coli* S30 extract contains endogenous components such as proteases and nucleases, the stability of the mRNA-ribosome-polypeptide ternary complex is compromised due to mRNA and nascent polypeptide degradation. Furthermore, the difficulty of optimizing the system for ribosome display is exacerbated by incomplete information on components within a cell extract. To address this issue, some authors created a highly controllable cell-free translation system known as the protein synthesizing using recombinant elements (PURE) system, which uses only purified factors and enzymes and lacks nucleases and proteases (Ohashi et al., 2012, 2010; Shimizu et al., 2001, 2005; Shimizu & Ueda, 2010).

Biopanning, as in phage display, is a typical method for identifying ribosome-bound peptides with desirable features from a peptide library, in which the ribosome-peptide complexes are immobilized on a solid support, such as a column or a plate, and the target molecule of interest is added. The ribosome-bound peptides that bind to the target molecule are then eluted and retrieved for further investigation (Chen et al., 2010; Marr et al., 2011). The following is a high-level summary of the steps involved in phage display screening.

- *Library construction*: In vitro transcription is used to convert a DNA library containing a T7 promotor, the peptide sequence, and a C-terminal spacer lacking a stop codon into mRNA. Ribosomes in stoichiometric quantities translate the mRNA templates. The ribosomes run to the end of the mRNA molecule after attaching to the Shine-Dalgarno sequence upstream of the start codon. Because there is no stop codon, the peptide exits the ribosome and folds into its 3D structure with the help of the enzyme disulfide isomerase, but without releasing the ribosome. The complex of peptide, ribosome, and mRNA is formed.
- *Affinity selection*: The biotinylated target (for example, an overexpressed receptor's extracellular domain) is immobilized on streptavidin-coated magnetic beads.
- *Washing*: Nonspecific binders are removed in repeated washing steps after the target has been incubated with the ribosome complexes.
- *Elution*: An elution buffer containing EDTA is used to isolate the mRNA. The Mg2+ ions are trapped by EDTA, which destabilizes the ribosome complex. The peptide and mRNA are both released. It is also possible to elute competitively with the target molecule or biotin to select high-affinity binders.
- *Purification*: A commercial kit is used to purify the mRNA for use in reverse transcription.
- *Amplification and enrichment*: Because the resulting cDNA lacks the T7 promotor and spacer, another PCR is required. A PCR is carried out using overhang primers containing the T7 promoter and the spacer region. If that step is performed with an error-prone PCR, the diversity of the selected peptide cDNAs increases. It takes 3–6 rounds to find peptides with low nanomolar or subnanomolar affinity.

After the initial screening, the ribosome-bound peptides can be further characterized and optimized through additional rounds of selection, amplification, and mutagenesis to improve their properties, such as affinity or specificity. Finally, the selected ribosome-bound peptides can be recovered, and the corresponding mRNA templates can be reverse transcribed to obtain the DNA sequences encoding the peptides for further analysis and downstream applications (Marr et al., 2011).

Enzymatic screening is also used in ribose display systems to screen for peptides with specific enzymatic activities by subjecting ribosome-bound peptides to enzymatic reactions (He, 2002). If the target is an enzyme, for example, the ribosome-bound peptides can be incubated with a substrate, and the formation of a product can be used to determine enzymatic activity. DNA encoding the protein of interest is transcribed and translated in vitro, just like in biopanning. Because mRNA lacks a stop codon and translation is inhibited by high Mg2+ concentrations, stable ternary complexes of mRNA, ribosome, and tethered nascent protein form. These can be used directly in conjunction with a biotinylated suicide inhibitor for selection. Gel filtration removes excess inhibitor, and avidin-coated magnetic beads capture the labeled complexes. The selected complexes are destroyed to elute the mRNA after being washed to remove any untrapped ribosomal complexes. Finally, reverse transcription (RT) and polymerase chain reaction (PCR) are used to amplify the genetic information of the selected clones (Amstutz et al., 2002).

Ribosome display is a high-throughput and efficient method for identifying peptides with desired properties, and it has been widely used in drug discovery (FitzGerald, 2000; Groves & Osbourn, 2005; Rothe et al., 2006; Yan & Xu, 2006), protein engineering (He et al., 2012; Ohashi et al., 2007; Schimmele et al., 2005; Shimizu et al., 2006), and functional genomics (Bieberich et al., 2000; Ullman et al., 2011). It enables rapid screening and selection of large and diverse peptide libraries, making it an important tool for the development of peptide-based therapeutics, diagnostics, and research reagents (Kunamneni et al., 2020). Overall, ribosome display is a powerful technique in traditional peptide library screening approaches that allows for the identification of peptides with specific properties, and it has been widely used in various fields of biomedical research (Dreier & Plückthun, 2011). As a result, it is a valuable tool for peptide discovery and engineering. However, as with any screening technology, careful experimental design, optimization, and validation are required for reliable and reproducible results. Furthermore, unlike other screening methods, the ribosome display method necessitates specialized equipment and expertise, making it more difficult to implement in some research settings (He & Khan, 2005; Rothe et al., 2006). Nonetheless, ribosome display is a valuable tool for screening peptide libraries, and its continued advancement and application are likely to contribute to the discovery and development of novel peptide-based therapeutics and other biotechnological applications.

2.3 Cell-surface display

Since the introduction of recombinant DNA techniques in the late 1970s, various proteins have been successfully expressed in a variety of host organisms such as bacteria, yeasts, fungi, mammalian cells, and plants.

One of the most exciting developments in recent years has been the ability to directly target proteins on the cell surface of various host organisms. This ground-breaking technology is opening up new possibilities for applications such as bacterial vaccines, HTS of peptide and enzyme libraries, whole-cell sorbents, recombinant biocatalysts, and cell-based diagnostics (Chen & Georgiou, 2002). Cell-surface display is accomplished by expressing a heterologous peptide or protein of interest as a fusion protein with various anchoring motifs, which are typically cell-surface proteins or fragments of cell-surface proteins (carrier proteins). C-terminal fusion, N-terminal fusion, or sandwich fusion strategies can be considered depending on the characteristics of the passenger and carrier proteins (Lee et al., 2003). The key challenge is for the peptide to be exported from the cytoplasm and directed to the outside membrane, where it must be capable of crossing the membrane and anchoring to the exterior surface (Chen & Georgiou, 2002). The subsequent subsections will go over cell-surface display for peptide libraries in bacterial, yeast, and mammalian systems.

2.3.1 Bacterial display

Since the first reports about bacterial display of heterologous proteins in 1986 (Charbit et al., 1986; Freudl et al., 1986), a broad number of different display systems have been established for Gram-positive and Gram-negative bacteria (Jose, 2006). Bacterial cell surface display process involves fusing the peptide library to a bacterial surface protein, such as the outer membrane protein OmpA (Bessette et al., 2004; Daugherty et al., 1998; Earhart, 2000; Lång, 2000; Vahed et al., 2020) or the flagellar protein FlgE (Shen et al., 2017; Zheng et al., 2015). A chimeric protein comprising Lpp and OmpA is used to show recombinant proteins on the surface of *E. coli*. It is made up of the LPP region (amino acids 1–9 of LPP) and the OmpA transmembrane sections B3-B7, to which the recombinant passenger is C-terminally linked. It has been used successfully to express enzymes, single-chain antibody fragments (scFv), and binding domains on the surface of *E. coli*, although it appears to be sensitive to the passenger protein's vast secondary and tertiary structures (Georgiou et al., 1997). Surface proteins in Gram-positive bacteria are often covalently attached to the peptidoglycan cell wall through a C-terminal anchoring tail. The cell wall anchoring regions from different naturally occurring surface proteins, including SpA (*Staphylococcus aureus* protein A), M6 protein (*Streptococcus pyogenes*), and FnBPB (fibronectin-binding protein B from *S. aureus*), have been used for the successful display of enzymes, epitopes and binding domains at the surface of *Staphylococcus xylosus*, *Staphylococcus carnosus*, and *Lactobacillus lactis* (Wernerus & Stahl, 2004). Once the library is expressed on the surface of the bacteria can be screened against a target molecule of interest, such as a receptor or an antibody (Jose, 2006).

Affinity-based screening of cell surface display libraries generally necessitates the use of fluorescence-activated cell sorting (FACS), because magnetic selection (MACS) alone or panning processes result in avidity interactions that interfere with screening affinity (Daugherty, 2007).

One advantage of this technique is that it allows for the screening of large peptide libraries, as the number of bacteria in a single culture can be in the billions. Additionally, bacterial surface display enables the screening of peptides in their native conformation, which can be important for certain applications, such as peptide-based therapeutics.

Overall, bacterial cell surface display is a powerful tool for peptide library screening and has been successfully used for a wide range of applications, including the discovery of new drug candidates, the engineering of enzymes, and the development of biosensors.

2.3.2 Yeast display

The yeast cell wall is a complex structure made up of polysaccharides and glycoproteins that are connected to the plasma membrane (Bolanos-Barbosa et al., 2023; Stewart, 2017). To perform yeast cell surface display, first clone the gene encoding the desired peptide or protein into a yeast expression vector containing a surface display motif, such as α-agglutinin or the agglutinin-like sequence (ALS) from *Saccharomyces cerevisiae* (Hoyer & Cota, 2016; Teymennet-Ramírez et al., 2022). Aga2p, a subunit of the α-agglutinin protein complex in *S. cerevisiae*, is one of the most used yeast cell wall proteins for surface display (Benz et al., 2020; Kizerwetter et al., 2023). The surface display motif serves as an anchor for the peptide or protein of interest, allowing it to be displayed on the yeast cell membrane's outer surface (Mei et al., 2017). The gene encoding the protein or peptide of interest is fused to the gene encoding the Aga2p protein in *S. cerevisiae*, and the resulting fusion gene is inserted into a yeast expression vector, which can be either *S. cerevisiae* or *Pichia pastoris* (Dai et al., 2023). The vector also includes a selectable marker, such as an antibiotic resistance gene, to allow yeast cells that have been successfully transformed with the vector to be selected (Peng et al., 2022; Tominaga et al., 2021). Following transformation, the yeast cells are grown in a selective environment to ensure that only cells with the fusion protein on their surface survive. Once the yeast cells have expressed the desired peptide or protein on their surface, they can be tested against a target molecule (Vanella et al., 2022). Incubating yeast cells with the target molecule, such as an antibody or protein receptor, and then selecting for yeast cells that have bound to the target using FACS or magnetic-activated cell sorting (MACS) to isolate cells that bind to the target, is how this is typically done (Wellner et al., 2021). Following the selection process, the DNA of the chosen yeast cells is sequenced to identify the peptide sequences responsible for the binding activity. These peptides can then be characterized and optimized for use in a wide range of applications.

Yeast cell surface display outperforms other screening methods in terms of throughput, ease of use, and ability to screen large peptide libraries (Teymennet-Ramírez et al., 2022; Uchański et al., 2019). Yeast cells are also eukaryotic, which means they are more similar to human cells than bacterial or yeast cells. This is especially helpful when looking for peptides that interact with human proteins or receptors. Yeast cells can also undergo post-translational modifications, which can affect the function of some peptides (Feiran Li et al., 2022). However, there are some limitations to yeast cell surface display. One limitation is that yeast cells are not as genetically tractable as bacterial cells, which makes optimizing the expression and display of a peptide or protein of interest more difficult (Teymennet-Ramírez et al., 2022). Furthermore, yeast cells are not as scalable as bacterial cells, making large-scale screening more difficult (Teymennet-Ramírez et al., 2022).

2.3.3 Mammalian display

Mammalian cells, such as Chinese hamster ovary (CHO) cells or HEK293 cells, can also be used for cell surface display of peptides (Chin et al., 2019; Kiermer, 2005). In this approach, the peptide library is typically fused to a membrane protein, such as CD4 or a cell adhesion molecule, which is then expressed on the surface of mammalian cells. Peptide-displaying mammalian cells can be screened using FACS or other techniques (Wang, Zhang, et al., 2023; Yanakieva et al., 2020).

Mammalian cell surface display is a powerful technique for the screening of peptide libraries. This method involves the expression of a peptide library on the surface of mammalian cells, allowing for the selection of peptides with specific binding properties (Crook et al., 2017).

To achieve mammalian cell surface display, a library of peptides is typically fused to a membrane-spanning protein, such as the transmembrane domain of CD8 or the platelet-derived growth factor receptor. The fusion protein is then expressed on the surface of mammalian cells, such as HEK293T cells or CHO cells, using a viral vector or a transfection approach (Javanmardi et al., 2021; Lavoie et al., 2019; Park et al., 2023; Wang, Zhang, et al., 2023).

The resulting library-displaying cells can be screened for binding to a specific target using a variety of methods, such as FACS or MACS (Bruun et al., 2017; Lievens et al., 2004; Mair et al., 2019; Robertson et al., 2020). Positive cells are then isolated, and the displayed peptides can be identified using sequencing or mass spectrometry (MS).

Mammalian cell surface display has several advantages over other screening methods, such as bacterial display, including the ability to display post-translationally modified peptides and the ability to screen for peptides that require mammalian-specific post-translational modifications for their activity (Bowers et al., 2014; Chang et al., 2023; Horlick et al., 2013). Additionally, mammalian cell surface display allows for the screening of larger libraries and may be more amenable to HTS methods (Luo et al., 2023).

Overall, mammalian cell surface display is a powerful tool for the screening of peptide libraries and has wide applications in drug discovery, vaccine development, and protein engineering (Ho & Pastan, 2009; Horlick et al., 2013; Robertson et al., 2020).

Once the peptide library is displayed on the cell surface, it can be screened for various properties, such as binding affinity, specificity, or functionality, by exposing the cells to a target molecule or performing functional assays (Crook et al., 2017). Peptides with desired properties can then be selected and further characterized for their potential applications, such as in drug discovery, diagnostics, or biotechnology (Crook et al., 2020). Mammalian cell surface display in classical screening technologies for peptide libraries offers a versatile and powerful approach for the identification of peptides with specific properties or functions, and it has been widely used in various fields of research and applications (Kiermer, 2005). Overall, cell surface display provides a robust and high-throughput method for screening large peptide libraries and identifying peptides with desired properties. It is a valuable tool in peptide engineering, drug discovery, and other areas of molecular biology and biotechnology. By using classical screening technologies in combination with cell surface display, researchers can efficiently identify peptides with specific properties and potential applications in various fields (Crook et al., 2017). This approach has been widely used in drug discovery, diagnostics, and basic research and continues to be a valuable tool for peptide engineering and other applications.

2.4 Advantages and disadvantages of classical screening technologies

It is important to note that the benefits and drawbacks of traditional screening technologies may vary based on the exact technology or test employed, as well as the context of the screening application. It is critical to carefully assess the benefits and drawbacks of each screening technology in the context in which it will be used and to consult with trained healthcare experts for suitable screening recommendations. Newer, more advanced screening technologies that address some of the limitations of traditional screening methods may also be available, and it is critical to remain up to date on the newest advancements in the area. Overall, the screening technology used should be based on variables such as the precise screening goals, target population, available resources, and the quality of evidence supporting the accuracy and efficacy of the technology.

2.4.1 Advantages

- *Established and validated*: Classical screening technologies have been widely used for many years, and their methods have been thoroughly validated in many applications. They have a proven track record of reliability and accuracy.
- *Cost-effective*: Classical screening technologies are often more affordable compared to newer, cutting-edge screening technologies. They are generally simpler and require less sophisticated equipment, making them more accessible to a wider range of users.
- *Simplicity*: Classical screening technologies are often simple to use and do not require extensive training or specialized expertise to operate. They can be performed by trained personnel in a relatively straightforward manner.
- *Established reference ranges*: Classical screening technologies have well-defined reference ranges, which are established based on large populations and extensive data. This makes it easier to interpret results and make informed decisions based on the screening outcomes.
- *Wide availability*: Classical screening technologies are widely available in many laboratories and healthcare settings, making them easily accessible for routine screening tests in many areas.

2.4.2 Disadvantages

- *Limited sensitivity and specificity*: Classical screening technologies may have limitations in terms of sensitivity (ability to detect true positive cases) and specificity (ability to correctly identify true negative cases). False positive or false negative results can occur, leading to misdiagnosis or missed diagnoses.
- *Limited scope*: Classical screening technologies may only screen for a specific set of conditions or parameters and may not provide a comprehensive screening profile. They may not be suitable for detecting rare or emerging conditions or for complex screening scenarios.
- *Invasive or time-consuming*: Some classical screening technologies may require invasive procedures or sample collection methods, which can be uncomfortable or time-consuming for patients. This may also limit the frequency or feasibility of screening in certain populations.
- *Lack of automation*: Classical screening technologies may rely on manual processes, which can be labor-intensive and time-consuming. This may increase the risk of human error and impact the efficiency of the screening process.
- *Limited sensitivity to early detection*: Classical screening technologies may not be as sensitive in detecting early-stage or asymptomatic conditions, as they are often designed for identifying advanced disease states. This may result in delayed or missed diagnoses, leading to potential adverse health outcomes.

3 Emerging technologies for peptide library screening

3.1 Microfluidics-based screening

Microfluidic devices, commonly referred to as "labs on a chip," are small-scale systems that enable precise manipulation and control of fluids at the micrometer scale (Aquino et al., 2021; Rodríguez, 2023). These devices consist of channels, chambers, valves, and pumps that work together to handle fluids within a controlled environment (Abizanda-Campo et al., 2023).

In recent years, microfluidic devices have emerged as a powerful tool in the fields of chemistry, biology, and medicine (Mashiyama et al., 2024; Rahman et al., 2024). They offer the ability to integrate multiple processes, such as mixing, separation, detection, and analysis, onto a single platform (Pinnock & Daniel, 2022; Rodríguez, Andrade-Pérez, et al., 2023). This allows researchers to conduct experiments with greater precision, efficiency, throughput, and lower cost than traditional laboratory techniques (Cottet et al., 2024; Popa et al., 2023; Rodriguez et al., 2021).

One of the most promising applications of microfluidics is in the field of peptide synthesis and screening (Sato et al., 2022). Peptides are short chains of amino acids that play critical roles in biological processes such as signaling, immune response, and enzymatic catalysis (Fujita et al., 2023; Ullah et al., 2023). However, the traditional methods for synthesizing and characterizing peptides are time-consuming and labor-intensive, often requiring large amounts of reagents and solvents (Masui & Fuse, 2022).

Microfluidic devices can significantly reduce the time and amount of reagents required for peptide synthesis and screening by offering better control over reaction conditions, such as temperature, pH, and reagent concentrations (Ali et al., 2023; Chang et al., 2018). This leads to improved yield and purity of synthesized peptides and faster characterization of their properties. Microfluidic devices have the potential to accelerate the development of new drugs and therapies and improve our understanding of complex biological systems (Sarma et al., 2023; Stutzmann et al., 2023).

3.1.1 Microfluidic chips for peptide synthesis

Solution-phase synthesis

Peptides, short chains of amino acids linked by peptide bonds, are vital molecules in various fields of research, including medicine and biotechnology, and play significant roles in many cellular processes (Díaz-Gómez et al., 2024; Gallardo-Becerra et al., 2024). The term “peptide” was first coined in the early 1900s by Emil Fischer, a German chemist, to refer to small compounds that resembled proteins but had lower molecular weights (Mitchell, 2024). Since then, extensive scientific research has been conducted on peptides, and their synthesis has undergone significant developments (Suresh Babu, 2011).

In 1901, Fischer and Fourneau reported the synthesis of the first dipeptide, glycylglycine, obtained by partial hydrolysis of the diketopiperazine of glycine (Forster, 1920). This breakthrough marked the beginning of peptide synthesis, and subsequent advancements led to the synthesis of more complex peptides with diverse applications. In 1932, Bergmann and Zervas created the first reversible Nα-protecting group for peptide synthesis, the carbobenzoxy (Cbz) group (Bergmann & Zervas, 1932; Fields, 2001). This discovery paved the way for the development of other protecting groups, which are essential for efficient peptide synthesis.

The solution-phase technique for peptide synthesis has been one of the most widely used methods since the 1950s (Jaradat, 2018; Kimmerlin & Seebach, 2005; Masui & Fuse, 2022). This technique involves the use of amino acid derivatives temporarily protected by protecting groups, which are added one by one to a reaction vessel containing a coupling reagent and a base. The amino acids are deprotected and coupled by forming peptide bonds, resulting in the synthesis of the desired peptide (Hughes, 2010). Although the solution-phase technique has several advantages, such as simplicity and versatility, it also has limitations, including the need for excess reagents, difficulty in controlling reaction conditions, and the potential for racemization (Kim & McAlpine, 2013).

Recently, microfluidic devices have emerged as a promising approach to overcome the limitations of the solution-phase technique and improve the efficiency and precision of peptide synthesis (Masui & Fuse, 2022). Microfluidics technology allows the manipulation of fluids on a microscale using channels with dimensions ranging from a few micrometers to a few hundred micrometers. This enables precise control of reaction conditions, reduced required reagent volumes, faster reaction times, and improved yields.

One noteworthy example of this approach is the work conducted by Fuse et al. (2019). In their study, they reported on the successful elongation of peptide chains via mixed carbonic anhydride, which was made possible by utilizing microfluidic technology to facilitate the rapid mixing of an organic layer containing a protected amino acid or dipeptide with an aqueous layer containing an unprotected amino acid or dipeptide, thereby expediting the desired amidation. The researchers observed that their approach was successful in suppressing undesired racemization/epimerization, with a maximum observed racemization/epimerization rate of $\leq 0.4\%$ (Fuse et al., 2019).

Furthermore, the study demonstrated the effectiveness of this method in synthesizing various di-, tri-, and tetra-peptides in good to high yields. This is significant, as it marks the first report of achieving peptide chain elongation from unprotected amino acids without severe racemization, using inexpensive, nonexplosive, less wasteful, and less toxic reagents. The authors noted that the use of mixed carbonic anhydride as an activating reagent for unprotected amino acids holds considerable promise, as it can be readily generated in situ under mild conditions, utilizing inexpensive, nonexplosive, less wasteful, and less toxic reagents. Moreover, any generated CO_2 and alcohol can be easily removed. While previous studies have explored the use of mixed carbonic anhydride for amidation with free amino acids, the suppression of undesired epimerization during the synthesis of peptides comprising three or more amino acids has proved challenging.

Fuse et al.'s work showcases the potential of microfluidic technology to facilitate peptide chain elongation using unprotected amino acids (Fuse et al., 2019). This approach demonstrates notable advantages over traditional peptide synthesis methods, as it eliminates the need for deprotection steps, reduces waste, and employs low-cost reagents. This promising advancement in peptide synthesis may have far-reaching implications in the pharmaceutical industry, enabling the development of more cost-effective and efficient methods for synthesizing peptides for therapeutic purposes.

Solid phase synthesis

Solid phase peptide synthesis (SPPS) is a powerful method for synthesizing peptides with high yield and purity. In SPPS, peptides are synthesized on a solid support, typically resin beads, that are functionalized with a protected amino acid (Goyal & Goyal, 2023; Guzmán et al., 2023). The protected amino acid is then deprotected and activated to allow the coupling of the next protected amino acid to form a peptide bond. This process is repeated until the desired peptide sequence is obtained (Goyal & Goyal, 2023). The origin of this revolutionary technique can be traced back to the groundbreaking work of R. B. Merrifield in the 1960s (Jaradat, 2018; Mitchell, 2008). Merrifield developed the first SPPS strategy and was awarded the Nobel Prize in Chemistry in 1984 for his achievement. Since then, SPPS has become an essential technique in peptide synthesis due to its simplicity and efficiency (Kaiser, 1984).

SPPS possesses several advantages over solution-phase peptide synthesis. One significant advantage is its ease of use, which allows for a more straightforward procedure compared to solution-phase synthesis (Coin et al., 2007; Palomo, 2014). Another advantage is the acceleration of the overall process, which leads to a quicker production of peptides (Palomo, 2014). Finally, SPPS can achieve good yields of purified products due to its efficient solid support (Mueller et al., 2020).

However, traditional SPPS also has several inherent disadvantages, including long reaction periods, excessive reagent consumption, and expensive apparatus (Mueller et al., 2020). These limitations have been addressed with the development of microfluidic systems for SPPS (Wang et al., 2011). These systems provide a high-efficiency, miniaturized, environment-friendly, and low-cost platform for SPPS. Microfluidic SPPS allows the continuous introduction of fresh reagents, usage of conventional reactants, simplicity of synthesis apparatus, low cost, and reutilization of the chips (Masui & Fuse, 2022). Furthermore, microfluidic SPPS offers several other advantages, such as the ability to perform multistep and cyclic SPPS, on-chip cleavage, and the production of peptides with different sequences and lengths simultaneously with good yields and purities (Masui & Fuse, 2022).

To perform SPPS within microfluidic devices, an FMOC strategy is often employed (Bhardwaj et al., 2014). The FMOC strategy involves the temporary protection of the N-terminus of the amino acid with the fluorenyl methoxycarbonyl fluorenyl methoxycarbonyl (FMOC) group, which is removed in each cycle to allow the addition of the next amino acid (Bhardwaj et al., 2014). This stepwise assembly of the peptide chain on a solid support, such as a resin or microbead, enables the synthesis of a fully assembled peptide chain, which can then be cleaved from the solid support and purified. The advantages of the FMOC strategy include the high purity of the final product, ease of purification, and versatility in the types of amino acids that can be incorporated (Masui & Fuse, 2022).

An example of this was the device developed by Wang et al. where the study involved an integrated multichannel SPPS that was carried out on a radial microfluidic system (Wang et al., 2011). Six peptides with strong affinities were synthesized on a hexachannel chip. In this process, the synthesis process began by attaching the C-terminal amino acid of the target peptide to hydroxymethyl microbeads through esterification of the functionalized linker and the symmetrical anhydride of Fmoc-protected amino acid in the presence of DMAP, using the Fmoc strategy. Peptide chain elongation was carried out in a circulatory reaction comprising deprotection, activation, coupling, and washing in each cycle. Deprotection reagents were delivered at a flow rate of $12\,mL\,min^{-1}$ for 5 min, and amino acid reagents with coupling solution were added from the amino acid feeding reservoirs. A pneumatic injector joined with the inlet of the diffluent chip was driven by the syringe pump at a flow rate of $12\,mL\,min^{-1}$ for 20 min. Six distributed powers from the outlets of the different chips infused into the reservoirs to drive the reagents. A cleavage reagent was injected through the axis inlet for 30 min at a flow rate of $12\,mL\,min^{-1}$, and a washing reagent DMF was injected through the axis inlet for 6 min at a flow rate of $300\,mL\,min^{-1}$. After peptide elongation, on-chip peptide cleavage was carried out. Continuous-flow cleavage reagent (97.5% TFA, 2.5% H_2O) was injected through the two inlets for 30 min at a flow rate of $2\,mL\,min^{-1}$, and the peptide solution was received from the outlet. The synthesized leucine-enkephalin with a sequence of Tyr-Gly-Gly-Phe-Leu was used as a model compound for the study. The level of attachment of the first amino acid was estimated by measuring the Fmoc groups cleaved from the resin using UV spectrometry. The prepared Fmoc-amino acid attached resins were then injected into the reaction chamber and trapped by the cofferdam-fence structure.

3.1.2 *Microfluidic chips for peptide library screening*

In recent years, the importance of screening techniques for generating peptide libraries has increased significantly due to their ability to identify peptides with specific properties for various applications (Jenne et al., 2023; Wang, Wang, et al., 2023). However, traditional screening methods have limitations such as high costs and long screening times. To overcome these challenges, new screening techniques have been developed that are more efficient, high-throughput, and automated (Clayton, 2021). Among these techniques, microfluidic devices have emerged as a promising alternative for peptide library screening due to their versatility and ability to precisely control reaction conditions (Saberi-Bosari et al., 2019). Microfluidics is the study of the behavior and manipulation of fluids in small volumes within microfabricated channels (Rodríguez, Báez-Suárez, et al., 2024; Rodríguez, Guzmán-Sastoque, et al., 2024). Microfluidic devices offer precise control over reaction conditions, enabling the screening of large numbers of peptides in parallel with minimal sample consumption. Moreover, microfluidic devices can be designed to enable various screening strategies, including Biopanning, making them an attractive option for peptide library screening.

Biopanning

The Biopanning technique is a valuable tool for screening peptides based on their affinity for a specific target of interest (Ma et al., 2023; Wang et al., 2021). This method involves exposing the molecules of interest, such as proteins or antibodies, to the target, allowing those with a high affinity to bind to the target while those without affinity are washed away. Finally, the strongly adhered peptides are collected for further analysis (Li, Pu, et al., 2023). However, the traditional method of biopanning is often prone to errors during the washing stage. This is due to the difficulty of achieving homogeneous shear stress, as the shear stress varies depending on the location of the material in an orbital shaker. Moreover, the traditional method makes it challenging to control and quantify the shear stress generated during the washing process.

To overcome these difficulties, microfluidic devices have been developed in recent years, which provide a more standardized and controlled approach to biopanning (Han et al., 2023; Wang et al., 2021). Among these devices, two major trends are the syringe pump-based biopanning method microfluidic devices, and centrifugal microfluidic devices. Of these, the centrifugal microfluidic devices have shown advantages in allowing more precise and controllable application of shear stress. This was demonstrated in the work of Han et al., who developed a microfluidic biopanning platform that employs centrifugal force to generate controlled high shear stress, enabling efficient screening of microbial surface display libraries for cells or peptides with high affinity toward a specific target material (Han et al., 2021).

In their study, Han et al. used gold and indium tin oxide (ITO), which are commonly used in electronics applications, as target materials (Han et al., 2021). The microfluidic platform facilitated the discovery of new functional materials for biosensing applications. The authors conducted experiments to determine the optimum shear stress condition using four *E. coli* strains with different affinities toward gold. This was used to isolate cells with novel peptide sequences that selectively bind to gold or ITO from the bacterial display library. The discovered peptides underwent further analysis using material surface binding assays, flow cytometry, and DNA sequencing.

The microfluidic platform, which allows for the precise control of high shear stress, is intended to be straightforward to construct and operate, offering a significant advantage over conventional biopanning and syringe pump-based microfluidic biopanning approaches. To effectively sort the eCPX 3.0 display library and induce peptide display, the authors employed a sorting procedure. Negative sorting was conducted off-chip on a glass substrate coated on both sides with a thin layer of the target material. Following this, the microfluidic platform, which relies on centrifugal force, was used to perform on-chip cell binding and wash steps. The sorted cells were then regrown or plated on agar plates to obtain single strains, and this process was repeated for several sorting rounds. In the final round, the sorted cells were subjected to DNA sequencing to identify novel peptide sequences that exhibit strong and specific binding to the target material.

As such, it allows for more efficient isolation of microorganisms that are strong binders to target materials of interest. The authors screened the eCPX 3.0 library and identified 208 isolates that specifically bind to gold, ITO, or both and identified specific peptides displayed on the surface of these cells that correlate with their binding to target materials.

This microfluidic platform developed by Han et al. demonstrates its versatility and potential for application beyond the specific materials studied. It can be used to screen other types of display libraries against any planar material format on which the centrifugal microfluidic channel can be placed, making it a powerful tool for the discovery and development of novel functional materials for biosensing and other applications.

Combinatorial peptide

Combinatorial chemistry has revolutionized drug discovery since Geysen et al.'s pioneering work in 1984, which introduced a novel method for synthesizing and analyzing the antigenicity of hexapeptides (Geysen et al., 1984). This method utilized a 96-well plate multipin system, enabling the rapid screening of a vast number of hexapeptides to identify highly antigenic peptides for developing vaccines against FMDV. Frank's modification of this method in 1992 (Frank, 1992), involving the chemical assembly of short peptide chains on a paper membrane, further enhanced the efficiency and scalability of peptide synthesis and screening. Since then, these methods have undergone continuous improvement, culminating in the development of highly efficient microarray methods for the immobilization and screening of compound libraries, such as the Microfluidic Combinatorial Peptide Microarray Synthesis (Monti et al., 2023; Puentes et al., 2020).

Recently, Li et al. introduced a new approach to producing combinatorial peptide microarrays using a microfluidic impact printing (MI Printing) technique (Li, Zhao, et al., 2019). In their study, the authors compared the MI Printing method with the established microfluidic pneumatic printing platform, which produced peptide microarrays with high repeatability.

The MI Printing technique utilizes a detachable/disposable polydimethylsiloxane (PDMS) cartridge mounted on a commercial dot matrix printer head with five pins. During printing, the pins squeeze the cartridge membrane to generate a droplet. Unlike traditional inkjet printing techniques that employ expensive piezoelectric or thermal electric materials integrated with the cartridge, the MI Printing technique utilizes a low-cost PDMS material, resulting in a disposable cartridge design that reduces costs and avoids contamination. The cartridge's geometry enables the production of small-size (80 μm) and multiplexed (5 channels per cartridge) droplet arrays, showcasing its ability to generate combinatorial peptide libraries. The MI Printing system's low dead volume (less than 0.05 μL) makes it highly desirable for peptide synthesis research, considering the high cost of some amino acids.

To demonstrate the applicability of the MI Printing technique for biomedical purposes, Li et al. synthesized a combinatorial tetrapeptide library with 625 permutations, which was screened with Jurkat lymphoid malignant T-cells for α4β1 integrin targeting. To achieve a high yield from peptide synthesis, they fabricated an amino-functionalized polyethylene glycol (PEG) microdisc array as the solid support for peptide synthesis on a glass slide support. The PDMS frame was then attached to prevent crosstalk.

Amino acids were dispensed into the PDMS wells using the MI printer and coupled to the microdisc array. Standard Fmoc-protected chemistry was employed, where the next Fmoc-amino acid was coupled after the protecting group was removed. The coupling cycle was repeated until the last amino acid was coupled. The entire synthesis process took less than 2 h for each coupling cycle, including printing (20 min, varying on library size), Fmoc deprotection (10 min), and washing (10 min). The coupling yield was verified by synthesizing a tetrapeptide, Nle-Asp-Phe-Glu, and sequencing the resulting microdiscs on a Procise protein sequencer. The sequencing results showed little impurities, demonstrating the high coupling efficiency of the MI Printing synthesis method.

To target α4β1 integrin, Li et al. designed and synthesized a tetrapeptide library with the N-terminus capped with 4-[(N′-2-methylphenyl)ureido]phenylacetic acid (UPA). The library referred to as UPA-X1X2X3X4, where X1, X2, X3, and X4 represent the last, third, second, and first amino acids coupled to the substrate.

3.1.3 Droplet-based microfluidics

The synthesis of droplets and double emulsions through the utilization of microfluidic devices stands as a pioneering technique at the confluence of experimental biology and chemical engineering (Shang et al., 2024; Yazdanparast et al., 2023). This emergent technology provides a distinct advantage by significantly expediting the execution of an array of diverse tests, thereby engendering a paradigm shift in the methodologies employed for the screening and generation of peptide libraries (Campaña et al., 2020; Rodríguez, Guzmán-Sastoque, et al., 2023; Wu, Wu, et al., 2023).

Droplets, within this context, represent infinitesimal, distinct liquid entities confined within a surrounding fluid, while double emulsions encompass a hierarchical arrangement, where droplets reside within droplets. These microscale compartments serve as exceptionally compact, tightly controlled reaction vessels, akin to micrometer-scale reactors. The precision and versatility inherent to microfluidic technology are instrumental in enabling the exacting manipulation and confinement of chemical reactions within these minute, controlled environments.

The utilization of droplet and double emulsion systems bequeaths a distinctive advantage to the field of peptide screening (Nuti et al., 2022; Yaginuma et al., 2019). By leveraging these systems, researchers are empowered to conduct a profusion of tests with an optimal allocation of resources, rendering the process highly efficient and cost-effective (Weng

& Spoonamore, 2019). This technological advancement equips scientists with the capacity to assemble expansive peptide libraries with unprecedented celerity and precision, thereby ushering in a transformative era in the domain of peptide research and discovery.

Fluorescent detection is a widely employed technique for screening peptides within droplets or double emulsions (Nuti et al., 2022). This method relies on the absorption of light by a fluorescent molecule, followed by its emission at a longer wavelength. To enable peptide detection, a suitable fluorophore is affixed to the target peptide (Pourmasoumi et al., 2023a, 2023b). The selection of an appropriate fluorophore is crucial, considering factors like spectral properties, photostability, and labeling efficiency (Trinh et al., 2023). Spectral properties encompass characteristics such as excitation and emission wavelengths, which should align with the instrument's detection capabilities. Photostability is essential to prevent fluorophore degradation or bleaching during the detection process. Labeling efficiency refers to the percentage of labeled peptides in the sample.

This methodology has facilitated the generation of various peptidic libraries, as exemplified by the work of Yaginuma et al. (2019), who focused on peptides targeting G-protein coupled receptors (GPCRs). GPCRs are intricately associated with a multitude of diseases and play a pivotal role in the realm of drug discovery. To pursue this endeavor, Yaginuma et al. devised a functional cell-based assay methodology, harnessing human reporter cells designed to express a specific GPCR (hGLP1R), yeast cells secreting either hGLP1R agonists or randomized peptide ligands, and a microfluidic droplet system. This ingenious utilization of a droplet microfluidic system facilitated the cocultivation of reporter and yeast cells, thereby enabling HTS of functionally active ligands. Yaginuma et al. took the initiative to engineer a reporter human cell line that emits fluorescence upon activation of hGLP1R, and they replaced the reporter protein to enable ligand activity evaluation through fluorescence. This alteration significantly enhanced the compatibility of their methodology with fluorescence-activated sorting, consequently augmenting the throughput for functional ligand identification. Notably, their study demonstrated that the cocultivation of yeast and mammalian cells within droplets resulted in an amplified reporter signal, even when the mammalian reporter cells were nonviable. This enabled the successful identification of functional peptide ligands, including one with superior activity compared to the original peptide sequence. This finding strongly suggests that their devised system holds promise in expediting the discovery of functional peptide ligands targeting GPCRs. The integration of a droplet microfluidic device, a key feature of their methodology, proved to be highly advantageous. This device enabled the concurrent assessment of a substantial number of ligand-secreting yeast mutants, thereby markedly enhancing throughput when compared to prior methodologies. Furthermore, Yaginuma et al. meticulously isolated yeast cells from the droplets and convincingly demonstrated that the identified peptide variants remained unaffected by post-translational modifications.

3.1.4 One-bead-one-compound (OBOC)

Combinatorial peptide libraries produced through the one-bead-one-compound (OBOC) approach are a highly effective and dependable method for generating a wide range of unique compounds supported on solid phases (Wang, Wei, Zhang, et al., 2014; Yan, Wang, Wang, 2023; Yan, Wang, Zhang, et al., 2023). This technique offers numerous advantages over biological peptide library techniques, including enhanced diversity, reproducibility, and ease of synthesis. The OBOC library concept, introduced by Lam et al., is based on the notion that each bead in the combinatorial library features a unique type of compound, with only one compound present on each bead, despite there being up to 1013 copies of the same compound on a single 100 μm diameter bead (Lam et al., 1991).

Despite the benefits of OBOC libraries, traditional screening methods for these libraries can be time-consuming and laborious. HTS methods have been developed to expedite the efficient identification and sorting of high-affinity ligands, increasing the probability of finding compounds with desirable properties (Yan, Wang, Zhang, et al., 2023). One approach utilized to improve the screening process is the use of microfluidics devices, which manipulate small volumes of fluids in microscale channels, allowing for HTS by running multiple samples simultaneously (Wang, Wei, Wang, et al., 2014). These devices have been utilized to streamline the screening of OBOC libraries by facilitating the efficient identification and sorting of high-affinity ligands (Cho et al., 2016). They offer several advantages, including precise control of flow rates, low sample volumes, and the ability to perform multiplexed assays (Li et al., 2018).

Furthermore, microfluidic devices provide a means of carrying out reactions on a small scale, which is critical when dealing with limited sample availability in drug discovery. Additionally, microfluidics offers a platform for performing assays in a highly controlled and reproducible environment, resulting in high assay sensitivity and accuracy.

The utilization of magnetic nanoparticles is one of the HTS approaches developed to overcome the cumbersome and demanding nature of conventional screening methods for OBOC libraries. Cho et al. presented a microfluidic device made entirely out of PDMS that offers a simple, low-cost, and reliable means of screening large OBOC libraries with high-throughput (Cho et al., 2016). The device employs four 21-gauge dispensing needles fitted into four holes on the lower PDMS channel, through which the library beads are continuously dispensed into the channel. A magnet positioned underneath the channel magnetizes the magnetic particles conjugated with the target protein, resulting in the magnetization of high-affinity beads. The positive hit population is then isolated through magnetic sorting using a micropipette.

The specific magnetic particles utilized in the process played a critical role in the success of the device and the accuracy of the sorting process. The device's accuracy of the sorting process was reported to be >98%, enabling comprehensive retrieval of beads for multiple additional rounds of screening. The synthetic nature of OBOC libraries allows for the incorporation of various building blocks in ligand production, resulting in ligands that are more resistant to proteolytic degradation. The HTS approaches developed, including the microfluidic device designed by Cho et al., have immense potential in drug discovery and development (Cho et al., 2016).

3.2 Other emerging technologies

3.2.1 Nanoparticle-based systems

Nanoparticle-based platforms are emerging as a promising approach for screening peptides due to their high sensitivity, selectivity, and compatibility with various detection methods. In this section, we will discuss four types of nanoparticle-based platforms used for peptide screening, including plasmonic nanostructures, QDs, magnetic nanoparticles, and carbon nanotubes, along with their advantages and disadvantages.

Gold nanoparticles (AuNPs)

Gold nanoparticles (AuNPs) are employed as versatile nanomaterials in various scientific applications due to their unique physicochemical properties, including their tunable size, shape, and surface properties (Atthar et al., 2023; Wang et al., 2024). They offer a promising platform for screening peptides because of their distinct characteristics. The use of AuNPs as a screening tool is advantageous due to their biocompatibility, ease of functionalization, and the potential for controlled material synthesis under environmentally friendly conditions (Kim et al., 2023; Li, Huang, et al., 2023). These nanoparticles possess exceptional optical and plasmonic properties, making them a valuable resource for studying interactions between peptides and solid substrates (Ferrari, 2023; Kumar & Lim, 2023; Selmani et al., 2023).

In the context of peptide screening, AuNPs can be integrated into various techniques that capitalize on their properties. Surface-enhanced Raman scattering (SERS) and localized surface plasmon resonance (LSPR) sensing are notable methods employed in peptide detection and screening. SERS entails amplifying Raman scattering signals of molecules in proximity to AuNPs, while LSPR sensing detects changes in the refractive index of the surrounding medium induced by the AuNPs. Both techniques can achieve high sensitivity and selectivity in peptide detection, while offering opportunities for further functionalization with peptides or antibodies. However, it is crucial to acknowledge that these techniques are not without challenges, as they may have limited quantification capabilities and issues related to result reproducibility, standardization, and optimization for diverse applications. Furthermore, the cost and availability of some plasmonic materials can pose practical obstacles.

The study conducted by Tanaka et al., as detailed in their research article, explores the application of AuNPs in peptide screening (Tanaka et al., 2017). They implemented a comprehensive strategy that integrated peptide arrays and identified peptides with AuNP-binding properties. The principle underlying their methodology involved designing an array informed by amino acid frequencies found in high-affinity binding peptides derived from empirical results. This approach not only enabled the identification of peptides with binding capabilities but also facilitated the discovery of approximately 1800 peptides exhibiting diverse binding activities.

The results of the study revealed a range of binding activities, with some peptides exhibiting high-affinity binding to AuNPs, while others displayed weaker or null-affinity binding. Notably, among the top-ranked peptide sequences, no single consensus sequence was observed; however, a common characteristic was the presence of one or more tryptophan residues. Interestingly, peptides containing multiple tryptophan residues with a noncontinuous configuration demonstrated higher binding affinities to AuNPs. These observations highlighted the significance of tryptophan residues in mediating interactions between peptides and AuNP surfaces, primarily through π-electron interactions. The study also identified motifs, such as W[*Xn*]W, H[*Xn*]W, and W[*Xn*]H, which were found to be associated with strong binding to AuNPs.

Quantum dots

Within peptide library screening, conventional fluorescence-based techniques encounter persistent challenges, especially when interrogating bead-bound combinatorial libraries to identify protein ligands (Garske & Denu, 2006; Zhou & Ghosh, 2007). The complexity arises from the interplay between fluorescent markers, like Texas Red, and the inherent resin autofluorescence (Olivos et al., 2003). This interaction often compromises sensitivity, leading to inconclusive results and necessitating meticulous visual inspection (Olivos et al., 2003). Consequently, the screening process becomes labor-intensive and prone to false positives.

The emergence of QDs has sparked a transformative shift in this landscape (Suwatthanarak et al., 2020). Initially noted for their unique attributes—heightened quantum yield, enhanced photochemical stability, and the ability to absorb broad-spectrum energy while emitting specific, sharp peaks—QDs emerged as intriguing alternatives to traditional organic fluorophores. The seminal contributions of pioneers like Moungi G. Bawendi, Louis E. Brus, and Alexei I. Ekimov in understanding and harnessing QD properties revolutionized nanomaterials research.

In recognition of their groundbreaking contributions to QDs, the 2023 Nobel Prize in Chemistry honored Moungi G. Bawendi, Louis E. Brus, and Alexei I. Ekimov (Yu & Schanze, 2023). Their profound insights and discoveries laid the foundation for the diverse applications of QDs across scientific disciplines. Although their work predominantly focused on the fundamental understanding and utilization of QD properties in nanomaterials, their discoveries had significant ramifications across various fields, including the intriguing potential for visualizing biomolecular interactions, such as detecting biotinylated proteins on beads. Notably, the reduced susceptibility of QDs to resin autofluorescence rendered them particularly promising in biomolecular visualization techniques.

In their groundbreaking research, Olivos et al. showcased the potency of QDs as a visual tool for screening bead-bound combinatorial libraries (Olivos et al., 2003). Their study meticulously compared QDs with established organic fluorophores like Texas Red. The results were striking—QDs exhibited significantly enhanced contrast, enabling clearer differentiation between positive and negative binding events. This heightened contrast streamlined the visual screening process, facilitating rapid and precise identification of ‘hits’ within the libraries.

Olivos et al.’s innovative work, building upon foundational research in nanomaterials such as that of Bawendi, Brus, Ekimov, and others, highlights QDs’ pivotal role in overcoming traditional fluorophore limitations in peptide library screening (Olivos et al., 2003). Leveraging QDs’ heightened sensitivity and contrast properties, they revolutionized visualization methods, efficiently identifying potential protein ligands within combinatorial libraries.

Magnetic nanoparticles

The incorporation of magnetic nanoparticles into peptide screening methodologies signifies a groundbreaking advancement in nanotechnology’s impact on this field (Dietrich et al., 2023). Among these nanoparticles, magnetite—also recognized as iron (II, III) oxide or Fe3O4—stands out for its diverse applications across industries, particularly inefficiently concentrating samples via pull-down techniques (Ortegón et al., 2022; Sojková et al., 2023). Its widespread use in biotechnology underscores its pivotal role in precisely isolating biomolecules (Tanaka et al., 2008).

In the realm of screening peptide libraries, magnetic nanoparticles offer a host of advantages. Their adaptable surfaces, tailored through diverse linker functionalizations (Guzmán-Sastoque et al., 2024), facilitate customized interactions with peptides, simplifying selective isolation within complex mixtures (You et al., 2016). This adaptability heralds a significant evolution in peptide screening methodologies, paving the way for advanced strides in biotechnology and biomedical research.

An exemplary demonstration of the potential of magnetic nanoparticles in peptide screening is evident in the work of Furukawa et al. Their study employed thermoresponsive magnetic nanoparticles for affinity selection from yeast cell surface display libraries (Furukawa et al., 2003). These particles, created by coating magnetite nanoparticles with a polymer blend comprising N-isopropyl acrylamide (NIPAM) and a biotin derivative, acquired crucial temperature-sensitive properties. Their methodology intricately relied on specific protein-protein interactions. Biotin, recognized for its high affinity to avidin, played a central role. The introduction of biotinylated antigoat IgG is selectively bound to target cells displaying the ZZ domain, forming a complex upon the addition of avidin to the cell suspension. The ingenious design facilitated a temperature-triggered aggregation process, wherein the nanoparticles underwent phase transition beyond their lower critical solution temperature (LCST), resulting in the insolubility of NIPAM. This induced aggregation or flocculation of the magnetic nanoparticles carrying the captured target cells. Leveraging this temperature-induced aggregation, a magnetic field efficiently separated the aggregated nanoparticles, swiftly isolating the enriched target cells from the solution. This precise magnetic separation technique enabled rapid and accurate extraction of specific cells of interest. This iterative process, involving enrichment, isolation, and amplification of captured cells through cultivation, significantly enhanced the efficiency and specificity of the screening methodology. Furukawa et al.’s innovative approach exemplifies how such strategies streamline peptide screening and hold immense promise for unraveling complex protein-protein interactions and expediting the isolation of novel biomolecules (Furukawa et al., 2003).

Carbon nanotubes

Single-walled carbon nanotubes (SWNTs) possess distinct structural attributes that confer unparalleled properties, surpassing those of conventional materials. Emerging from their nanoscale framework and extraordinary physical traits, these characteristics position SWNTs as highly promising substrates, notably in the realm of peptide library screening applications. The cylindrical form of SWNTs, crafted from one-atom-thick graphene sheets rolled into minute tubes, imparts exceptional mechanical, electrical, and optical characteristics. Their elevated aspect ratio and notable tensile strength ensure mechanical resilience and adaptability, while their significant electrical conductivity originates from the dispersed π-electron system across their hexagonal lattice. Furthermore, their structure-sensitive optical traits, such as robust absorbance and photoluminescence within the near-infrared spectrum, identify them as optimal candidates for diverse optical applications. This amalgamation of attributes not only intensifies the scientific intrigue around SWNTs but also establishes them as versatile platforms for varied detection methodologies. Their inherent electrical conductivity facilitates the development of exceptionally sensitive electrical sensors, while their optical properties enable the creation of highly responsive optical probes. Moreover, the substantial surface area-to-volume ratio of SWNTs augments their potential for customization and integration with biomolecules, enabling tailored interactions with peptide libraries.

The synergy between SWNTs' unique structural properties and their multifunctional capabilities lays the groundwork for inventive approaches in peptide library screening. Leveraging these features, SWNTs hold promise in furnishing innovative, highly sensitive, and adaptable platforms for precise peptide identification, characterization, and manipulation. Illustrating this synergy is the research conducted by Krabacher et al., centering on peptide-driven recognition and bonding with SWNTs through biopanning methods (Krabacher et al., 2021). Conventional techniques, like phage display libraries, often confront limitations stemming from sequence biases and the loss of distinct clones during amplification cycles, thereby yielding false positive sequences. Addressing these challenges, a modified high-throughput next-generation sequencing (HTS) technique selectively isolates peptides binding specifically to various sources of SWNTs. A comparative assessment among three methodologies—traditional sequencing, HTS-based biopanning, and an innovative normalized HTS method (NHTS-PD)—underscored the superior efficacy of NHTS-PD in identifying and eliminating extraneous sequences, thereby revealing peptides exhibiting precise binding affinities to SWNTs. These identified peptides showcased selective affinities contingent upon SWNT diameter, a pivotal aspect for sensor applications. Characterization and comparison revealed the prevalence of aromatic amino acids, elucidating distinct preferences of peptides (P8 and P4) for diverse SWNT sources based on diameter and chirality. Validation of these findings occurred through dispersion experiments and functionalization with gold nanoparticles (AuNP). The peptides facilitated precise AuNP assembly on SWNTs, affirming specificity and the potential for multifunctional applications.

3.3 Advantages and disadvantages of emerging screening technologies

The landscape of peptide library screening has been fundamentally transformed by the advent of novel technological approaches, notably microfluidics-based systems and nanoparticle-based platforms. These innovations have catalyzed a paradigm shift in the field, offering enhanced precision and efficiency. However, alongside these advancements, they introduce a spectrum of challenges. This critical examination aims to elucidate these technologies' multifaceted advantages and inherent drawbacks, informed by current literature and empirical studies.

3.3.1 Advantages

Precision and control: Microfluidic devices exemplify the pinnacle of control in experimental conditions, manipulating fluids at micrometer scales. This precision is pivotal for achieving accurate and reproducible results, a notion supported by extensive research in the field. Their ability to fine-tune various parameters results in outcomes of unparalleled accuracy.

Efficiency and throughput: Integrating multiple processes onto a single platform, these technologies significantly enhance both efficiency and throughput. This is particularly salient in HTS contexts where large peptide libraries are processed concurrently. Such integration aligns with contemporary demands for expedited research methodologies.

Resource optimization: By necessitating lower reagent volumes, these technologies are both economically advantageous and environmentally sustainable. This aspect is crucial when handling expensive or limited-availability resources, aligning with a growing emphasis on sustainable research practices.

Sensitivity and selectivity: Nanoparticle-based systems, including QDs and magnetic nanoparticles, have been noted for their enhanced sensitivity and selectivity in peptide detection. This advancement is instrumental in identifying peptides with specific properties, a key objective in current peptide research.

Automation potential: The prospect of automation in these systems promises a reduction in manual labor and minimization of human error, factors that are increasingly critical in modern research environments. Automation is synonymous with consistency and reliability in experimental outcomes.

3.3.2 Disadvantages

Complexity and cost: The implementation of these technologies is not without its complexities and financial implications. The requisites of substantial investment and specialized operational knowledge may pose barriers to their widespread adoption in diverse laboratory settings.

Scalability challenges: Transitioning these technologies from small-scale research to larger production settings presents scalability challenges. Maintaining efficiency and precision during this transition is a critical concern in the broader application of these methods.

Maintenance and durability: The intricate nature of these systems often demands frequent maintenance, and their miniaturized components may offer less durability compared to traditional methods. These factors could contribute to increased long-term operational costs. Application Limitations: Certain technologies may not be universally applicable across various peptide synthesis and screening scenarios, necessitating careful tailoring to specific requirements. This limitation highlights the need for versatile and adaptable screening technologies.

Data complexity: The sophisticated nature of these screening methods often results in complex data sets that require advanced tools and expertise for effective analysis and interpretation. This complexity underscores the necessity for continued development in data processing capabilities.

Emerging technologies in peptide library screening, such as microfluidics and nanoparticle-based platforms, offer significant improvements in precision, efficiency, and sensitivity. However, they also introduce challenges including high initial costs, operational complexity, scalability issues, and the need for advanced data analysis capabilities. Addressing these challenges is crucial for the broader adoption and practical application of these technologies in the fields of biotechnology and pharmaceutical research. Future advancements should focus on enhancing user accessibility, scalability, and data management, thereby unlocking the full potential of these emerging screening technologies. This will ensure their alignment with the evolving demands of scientific research and their applicability in a variety of research contexts.

4 Approaches to monitoring screening in real time

Conventional approaches for peptide library screening, while useful and applicable in many cases, generally require time-consuming and laborious preparations, along with lengthy procedures that increase the likelihood of errors propagating throughout the entire screening methodology (Wang, Wei, Zhang, et al., 2014; Wei et al., 2021; Zhang et al., 2021). Some of the mentioned classical procedures possess elevated risks of producing false positive and false negative results, especially due to the lack of high-level chemical control and monitoring of the compounds and reactants used in the screening process (Wei et al., 2021; Zhang et al., 2021). Furthermore, some of these approaches also need excessive data processing by analysts, which further increases the time needed for a full screening of a complete peptide library (Wei et al., 2021). Thus, real-time approaches have been developed to overcome these issues and allow for peptide screening in a rapid and more controlled manner (Misra et al., 2015). Moreover, by using these real-time procedures, measuring sudden reactions or interactions between molecules also becomes possible, as more precise monitoring of the chemical compounds and intermediate species in the reaction can be done (Misra et al., 2015). Thereby, the next section will briefly summarize some of the most common approaches to monitoring screening in real time, including label-free methods, fluorescence-based methods, MS-based methods, and surface plasmon resonance-based methods.

4.1 Label-free detection methods

Label-free detection methods, as implied by their name, are screening methods that do not require the use of any radio-, enzyme-, or fluorescent labeling to report the presence of a specific ligand or molecule of interest in a determined sample (Cooper, 2003). Because of this, if compared to conventional fluorescent-based methods, label-free methods tend to be cheaper, have an easier sample preparation, be extremely sensitive, and provide more direct binding information, thus offering an excellent alternative to track real-time molecular interactions (Cooper, 2003; Cozzolino et al., 2020; Fang, 2010, 2014; Syahir et al., 2015). Specifically, these methods work by exploiting the biophysical properties (such as the charge, refractive index, and molecular weight) of ligands and proteins in a reaction to keep track of their molecular events and detect the presence or activity of any desired molecule or compound (Syahir et al., 2015). Therefore, in this process usually referred to as *biosensing*, these molecular interactions can be interpreted as electrical, mechanical, optical, or even thermal signals, which can be easily detectable without any label probes (Syahir et al., 2015). Moreover, because of limiting the use of labeling molecules, label-free methods do not occlude binding sites or interfere with reactions as could potentially happen with label-based methods, thus reducing possible false negatives (Cooper, 2003). The present section will now cover some common label-free methods that include optical biosensors, acoustic biosensors, and electrochemical biosensors.

Optical biosensors are devices that rely on light in all forms to measure and sense molecular events in real time without any labeling. Optical biosensors are the most common type of biosensor used too. They have multiple advantages such as being very sensitive, having small sizes, being highly specific, and being very cost-effective. Moreover, they can be easily coupled to new disciplines such as nanobiotechnology, microfluidics, and microrobotics, thus widening their range of use (Damborský et al., 2016). The principle of these biosensors is usually based on exploiting the interaction of an optical field with a biorecognition element, which when involved in a specific molecular reaction close to the sensor, usually causes changes in wavelengths or angles that can be tracked or sensed to give information about the occurred reaction (Damborský et al., 2016; Fang, 2010). Therefore, an optical biosensor is made of two main components: a biorecognition sensing element and a signal transducer system.

Specifically, the biorecognition sensing element is commonly a biological molecule or cell immobilized in a sensing plate; this biological element can go anywhere from an enzyme, antibody, or protein to molecules such as DNA or even whole cells. Thus, the purpose of this biorecognition sensing element is to react to a desired analyte (a protein, for example) and produce a signal (electromagnetic, chemical, etc.) that is proportional to the concentration of the analyte, thus sensing the affinity of the analyte to biorecognition element (Cooper, 2003; Damborský et al., 2016; Fang, 2010). However, this signal produced must be transduced and consequently processed to be analyzed and understood; that is why signal transducing systems are needed. As such, signal transducing systems are what makes each optical system different. There are several types of transducing systems, and each system creates a different type of optical biosensor. For example, the surface plasmon resonance detection method (which is described in detail in Section 4.4) is one of the most used optical biosensors: this transducer specifically detects changes in an incident light angle to determine whether a reaction is happening or not. Other types of optical biosensors include resonant waveguide grating (RWG), which measures changes in the refractive index of a metal surface where the biosensing elements are immobilized, ellipsometric biosensors which measure changes in the polarization of light, or even reflectometric interference spectroscopies (RIfS) which measure changes in the phase and amplitude of polarized light (Cooper, 2003; Damborský et al., 2016; Fang, 2010).

Acoustic (or piezoelectric) biosensors, in other words, are label-free biosensing devices that generally work by detecting changes in the physical properties of acoustic waves, such as their resonance frequency or resistance (Cooper, 2003; Fogel et al., 2016; Syahir et al., 2015). These sensors usually require the use of a piezoelectric material (their sensing element), which can be understood as a crystalline solid that lacks a symmetric center of inversion and, thus, possesses the ability to produce electrical charges on their surfaces when exposed to mechanical stress and vice versa (this is called a *piezoelectric* material) (Cooper, 2003; Fogel et al., 2016; Syahir et al., 2015; Uchino, 2017). Commonly, a quartz crystal is used as a piezoelectric material in these types of biosensors. Thus, if the quartz crystal surface is modified to be responsive to a certain ligand (for example, by immobilizing receptors that bind the analyte of interest in said surface), the quartz crystal becomes sensible to mass change. Therefore, when a binding event happens, the mass change in the surface of the quartz induces a mechanical response that, due to the piezoelectric effect, produces acoustical and electrical waves that can be measured (using sensitive microphones as transducers) to determine the course of a reaction or binding event of an analyte in real time (Cooper, 2003; Fogel et al., 2016; Syahir et al., 2015). Thereby, the produced acoustical waves account for the amount of ligand bound to the surface, meaning that the change in the resonant frequency of the acoustic wave is proportional to the quantity of ligand (peptide, for example) bound to the receptors in the quartz crystal (Cooper, 2003; Fogel et al., 2016).

This acoustic determination of binding events confers acoustical biosensors with a higher resolution and sensibility than optical biosensors. Aside from being sensible to small mass changes, acoustical biosensors can obtain further detailed information about a reaction as they are also extremely responsive to changes in viscoelastic properties of materials and charge changes of receptor-ligand complexes; moreover, they are even sensible to water-associated adsorbed proteins, which cannot be recognized by optical biosensors (Cooper, 2003; Fogel et al., 2016; Syahir et al., 2015). Finally, it is worth mentioning that acoustical biosensors are typically divided into two main categories that differ from each other according to the direction of propagation of the acoustic wave: surface acoustic wave (SAW) sensors or bulk acoustic wave (BAW) sensors (Fogel et al., 2016). Whether a sensor is cataloged as a SAW or BAW depends on the cut of the quartz crystal used; in fact, a crystal will vibrate and oscillate differently depending on how it is cut. Therefore, BAW sensors use an AT-cut crystal which makes the sound waves vibrate side-to-side, making the wave travel outwards to the bulk material; in other words, SAW sensors use an ST-cut crystal that makes the acoustic waves vibrate up and down, thus propagating the wave parallel to the quartz surface (Fogel et al., 2016). It is important to consider the mentioned sensor configurations, as the detection of the sound wave depends on how it is propagated. Moreover, these diverse configurations allow for thousands of different modifications to the baseline acoustic sensors, permitting the development of new biosensing platforms that may be sensible to specific material properties or reactions. For example, quartz crystal microbalance (QCM) is a type of BAW sensor that is especially sensible to very small changes in mass and, if modified to measure crystal oscillation dissipation, it can easily estimate the viscoelastic properties of materials (Fogel et al., 2016; Syahir et al., 2015). Therefore, it is evident that these biosensors are of great use to screen peptides, as they are extremely sensitive to ligands and can easily be used to determine binding affinity.

Finally, the last type of biosensor that will be mentioned is electrochemical biosensors. This group of sensing devices, as happened with optical and acoustical biosensors, also allows the monitoring of peptide screening in real-time without the use of any labeling; however, as its name implies, the signal transducing system used this time is based on electrical signals (like voltage, impedance, and current), and not optical signals or acoustical waves (Cho et al., 2020; Grieshaber et al., 2008; Wu, Liu, et al., 2023). Specifically, they work by coupling a biorecognition sensing element (such as an enzyme, protein, antigen, etc.) with an electrode or field-effect transistor that is extremely sensible to small changes in electrical signals. Thus, as many reactions involve electron movement and reorganization, by immobilizing said biorecognition elements in the surface of the electrode, reactions and binding interactions can be measured as a function of the change in the electric signals through time (Cho et al., 2020; Grieshaber et al., 2008; Wu, Liu, et al., 2023).

Although the obtained signal has a very small magnitude, it can be easily amplified with no further trouble by using physical or even chemical means, thus improving detection sensitivity (Wu, Liu, et al., 2023). Other advantages of electrochemical biosensors include their ability to be used in turbid biofluids without major background noise, excellent detection limits with small analyte volumes, and even low-cost production (Grieshaber et al., 2008). Moreover, these types of biosensors further have an excellent advantage over other sensing platforms, as they can be easily coupled with nanomaterials (such as carbon-based nanotubes, metallic nanoparticles, porous silica, or even organic polymers) for various applications (Cho et al., 2020). For example, the usage of nanoparticles in the coating of the main electrode augments the surface area of the electrode, making it more sensitive to electrical changes. What's more, nanotechnology can also be used to miniaturize these biosensors, therefore providing immense benefits in their application in chemical and biological contexts (Cho et al., 2020; Grieshaber et al., 2008; Wu, Liu, et al., 2023).

Finally, electrochemical biosensors can be further classified into various subcategories depending on the electrical signals that they measure (Grieshaber et al., 2008). For example, if a certain reaction generates a measurable current (resulting from the reduction or oxidation of a species), this current can be detected as it reduces the oxygen near a platinum electrode about an Ag/AgCl reference electrode; when this happens, the biosensors that measure the said current are called amperometric devices. Conversely, if a chemical reaction generates a potential or charge accumulation, which is measurable by comparing the potential to a reference electrode, the electrochemical biosensor used is called a potentiometric device. Moreover, if the reaction causes an alteration in the conductive properties of a medium between two reference nodes or electrodes, the device used is called a conductometric device. Finally, if a sensor measures the impedance caused by a reaction between two electrodes, the device is called an impedimetric device (Grieshaber et al., 2008).

Further biosensors include calorimetry, magnetic, gravimetric, and electric biosensors. All of them follow a similar configuration of a biorecognition sensing element coupled to a signal transducing system but differ from each other depending on the type of transducing system used and the physical or chemical variable measured (Cooper, 2003; Damborský et al., 2016). Almost all of them use enzymes, proteins, and antigens as sensing elements, which, yet again, make them suitable for monitoring reactions and, extremely usable for applications in real-time and label-free peptide screening monitoring.

4.2 Fluorescence-based detection methods

Fluorescence-based detection methods, also called label-based detection techniques, are defined as screening procedures that require the use of a labeling molecule (or *fluorophore*) to be able to register the progress of a reaction, quantify its products, and report its associated kinetics in real time (Demchenko, 2009; Ray et al., 2010). Specifically, label-based detection techniques fundament their *modus operandi* in binding radioisotopes, fluorescent dyes, epitope tags, or even nanoparticles to a specific analyte to sense and outline its chemical or physical changes when surrounded by a particular molecular environment or when involved in a specific molecular reaction. More commonly, this fluorescent molecule is bonded to the analyte along with a *quenching* molecule, which reduces the fluorescence intensity of the fluorophore, acting as a sort of deactivating molecule. Therefore, when exposed to said specific molecular event, the quenching molecule is separated from the fluorophore, thus increasing its fluorescence and allowing the quantification of the emission intensity when excited at a certain light wavelength that depends on the type of fluorophore (Demchenko, 2009; Larsen et al., 2021; Mirdha & Chakraborty, 2021; Ray et al., 2010).

Even though these techniques can affect target binding as the fluorophore may cause allosteric changes in the analyte or even generate a steric impediment, label-based techniques are, in theory, admissible in almost any molecular reaction; the only restriction is that, at least, one of the reactants should be able to be label-modified without changing or loosing binding affinity with its target (Demchenko, 2009). Apropos, it remains relevant also to mention that currently, there are hundreds of different ways to bind tags in analyte molecules. Moreover, there are also various ways to perform fluorescence-based assays. Categorization of fluorescence-based assays is completely dependent on various parameters and variables associated to the experiments such as the type of fluorophore used (for example, isotope labeling, chemiluminescent labeling, nanoparticle labeling, etc.), the chemistry behind the reaction (i.e., if the conjugation technique uses either amine-, thiol-, or click-reactive probes), and even the types of molecules involved in the reaction (like the Enzyme-Linked ImmunoSorbent Assay ELISA, which is named after the molecule used in the technique) (Demchenko, 2009; Larsen et al., 2021; Mirdha & Chakraborty, 2021; Ray et al., 2010; Syahir et al., 2015). However, there is one assay that is worth mentioning in greater detail: Förster/Fluorescence Resonance Energy Transfer (Demchenko, 2009; Mirdha & Chakraborty, 2021; Syahir et al., 2015).

Förster Resonance Energy Transfer (or FRET for short) is one of the most common label-based techniques used daily. In this method, two dye molecules interact with each other, exchanging energy due to a dipole-dipole effect and thus altering their abilities to emit and absorb light (Demchenko, 2009; Mirdha & Chakraborty, 2021; Syahir et al., 2015). Specifically, due to said dipole-dipole effect, one molecule (the *donor*), if excited, can transfer energy to the other molecule (the *acceptor*), which is therefore able to emit light, thanks to the received energy (Demchenko, 2009). This is called the *FRET effect* (thus the name of the technique) and depends completely on the distance between both donor and acceptor molecules. Therefore, if both molecules are close enough to each other (less than 10 nm), the donor molecule will be quenched, and the acceptor molecule will emit light at a certain wavelength that can be measured; furthermore, if the molecules separate considerably from each other, only the emission of the donor (which has a different wavelength than the acceptor molecule) will be observed (Demchenko, 2009). Thereby, this technique is useful in screening peptide libraries, as the acceptor and donor molecules can be placed in analyte and target molecules, thus being able to determine if a bonding interaction is taking place by just measuring the individual intensities of each dye molecule.

4.3 Mass spectrometry-based detection methods

MS is, in general, considered a technique that allows the identification of proteins based on their molecular mass and charge, typically after a tryptic digestion. More specifically, MS-based detection methods are considered to be label and crosslinking-free procedures that use these mentioned mass-charge relationships to quantitate proteins, monitor their levels, and characterize their interactions and chemical modifications (Gavriilidou et al., 2022; Syahir et al., 2015). Nowadays, there are several types of MS techniques, each one of them with a different resolution and throughput; however, the most relevant ones for peptide screening include electrospray ionization-MS (ESI-MS), matrix-assisted laser desorption ionization-MS (MALDI-MS), surface-enhanced laser desorption ionization-MS (SELDI-MS), and stable isotope ratio-MS (SIRMS) (Gavriilidou et al., 2022; Rohman & Wingfield, 2016; Syahir et al., 2015; von Neuhoff & Pich, 2005). This section will now focus on briefly describing these techniques and their typical characteristics.

ESI-MS (also called Native MS) is one of the most common and frequently used types of MS. Out of the various applications it has, ESI-MS is extremely useful for studying proteins, protein complexes, and interactions of proteins with small-molecule probes (Gavriilidou et al., 2022). Moreover, ESI-MS also allows the investigation of noncovalent interactions using just picomoles of a sample of interest, with a high resolution and a speed time of just milliseconds, thus being of great use to study protein-ligand interactions and their respective affinity (Gavriilidou et al., 2022). Furthermore, ESI-MS is capable of partially preserving the native quaternary and tertiary structure of proteins and may also maintain the noncovalent bonds between proteins and probes of interest (Gavriilidou et al., 2022). How does this technique work?

In general, an MS is based on the principle of fragmenting a molecule into several pieces or subunits which, having acquired different formal charges thanks to said break, become ionized and can now be separated and classified through the use of a magnetic field by their weight and their acquired charge (Amorim Madeira & Helena, 2012; Gross, 2017; Olshina & Sharon, 2016; Soares et al., 2012). This rupture can be done in various ways and, to ensure the ionization of each piece, the compounds are also usually subjected to collisions with hydrogen or other treatments. For example, the specific way ESI-MS does this rupture is by transferring a desired peptide sample from its buffer to a tryptic volatile solution (with the help of gel filtration, size exclusion chromatography (SEC) or dialysis) and consequently loading this solution (alongside with the probe of interest) into an electrospray ionization capillary (Gavriilidou et al., 2022). Consequently, this ESI capillary applies electricity to the sample and sprays the solvent peptides, evaporating them and turning this solvent into a gaseous-droplet phase (Gavriilidou et al., 2022; von Neuhoff & Pich, 2005). This process ionizes the peptides and generates various peptide fragments with different charge states, each with a specific mass-to-charge ratio. In this way, these ions, when subjected to the mentioned magnetic field, will collide at various points on a detection plate (depending on their mass/charge ratio) and thus generate a measurable signal that, consequently, can be graphed according to the intensity of the signal at any given point on the board (Amorim Madeira & Helena, 2012; Gross, 2017; Olshina & Sharon, 2016; Soares et al., 2012). This type of MS that uses electric fields for sample analysis is frequently called time-of-flight MS, so it is common to also call these techniques ESI-TOF-MS. In this order of ideas, a graph will be obtained which indicates the proportions of each fragment within the original compound, therefore making it possible to reconstruct not only the molecular weight of the parent compound, but even its original structure (Amorim Madeira & Helena, 2012; Gross, 2017; Olshina & Sharon, 2016; Soares et al., 2012). What's more, various MS can be done one after another to increase the resolution of the results; this process is called Tandem MS (or MS/MS) (Amorim Madeira & Helena, 2012; Soares et al., 2012). Therefore, this technique is of use in peptide screening because, when added together, the affinity of the peptide for a desired ligand can be easily determined just by looking at the resulting MS spectrum. In this way, the affinity between these compounds will be defined depending on the intensity of the resulting protein-ligand peak, compared to the intensity of the pack for the bare protein (Gavriilidou et al., 2022).

However, ESI-MS has one big disadvantage: it is not made for high-throughput assays and, if used along a SEC, it becomes a time-consuming process (Gavriilidou et al., 2022; Rohman & Wingfield, 2016). Therefore, new modifications to the MS method have been created to increase the throughput of this technique. Some of these modifications include making a screening cascade with a normal native-MS, where several wells are used allowing for probe multiplexing; separating peptide ions based on their charge and shape (not their mass) in a procedure called ion mobility mass spectrometry (IM-MS); modifying the electrospray platform to make it automated and consistent to improve reproducibility (chip-based nano-ESI-MS), or even using a solid (not gaseous) peptide sample, just like happens with the matrix-assisted laser desorption ionization-mass spectrometry (MALDI-MS) (Duncan & DeMarco, 2019; Gavriilidou et al., 2022; Rohman & Wingfield, 2016). This last technique, MALDI-MS (or MALDI-TOF-MS), is of particular interest because, as a consequence of using a solid sample, the need to employ transferring/separating techniques such as the mentioned SEC is eliminated, speeding up the sample analysis and allowing sample preparation to occur in parallel to sample study (Duncan & DeMarco, 2019; von Neuhoff & Pich, 2005).

MALDI-MS works as follows: first, a preprocessed sample of the protein of interest is mixed with a matrix solution (usually made of siapinic acid or α-cyano-4-hydroxycinnamic acid) and placed over a metallic plate (Duncan & DeMarco, 2019; Whelan et al., 2008). Thereafter, this mixture is left to dry and crystallize, so that it forms a solid composition. Next, the solid mixture is placed into the MS where a series of pulsed laser beams irradiate both the peptide and matrix molecules. Then, the matrix molecules absorb the laser's energy and, consequently, ionize themselves and the adjacent protein molecules, which are later segregated and separated by their mass/charge ratio in the MS in a similar way to that of the ESI-MS (Duncan & DeMarco, 2019; von Neuhoff & Pich, 2005). Thus, the resulting spectrum shows a similar intensity vs. m/z ratio plot, which serves as well to determine the affinity of a protein to a certain ligand just as explained before with ESI-MS.

Finally, two other MS techniques are commonly used for peptide screening and are considered to be high throughput: surface-enhanced laser desorption/ionization mass spectrometry (SELDI-MS) and stable isotope ratio mass spectrometry (SIRMS). SELDI-MS, for its part, is a modification of MALDI-MS. In this technique, a siapinic acid or α-cyano-4-hydroxycinnamic acid matrix is also used to facilitate the ionization of a protein sample. The substantial difference, though, between MALDI-MS and SELDI-MS resides in that SELDI-MS previously binds a sample of the protein of interest to a chip or array by affinity. In this way, the chip can be washed multiple times to ensure that no unbound protein or contaminant remains in the plate, thus leaving only the proteins that have an affinity for the ligands on the chip; this represents an advantage due to reduced sample impurity. Afterward, the mentioned matrix is placed over the protein patches and subsequently targeted by a laser just like in MALDI-MS (Syahir et al., 2015; Whelan et al., 2008). The result of this beam excitation is, as expected, the ionization of the samples and their classification due to their m/z ratios with the help of electric fields created by a TOF-MS (Syahir et al., 2015; Whelan et al., 2008). Therefore, affinity determination can be done, as explained before, with MALDI-MS and ESI-MS.

SIRMS, in other words, is yet another technique that also uses the m/z ratio to classify ionized protein fragments, as the other methods do. However, this method is particularly interesting as it focuses on the usage of stable isotopes of diverse molecular mass to tag proteins and classify them based on their final cumulative mass (von Neuhoff & Pich, 2005). This idea provides a higher sensitivity to affinity and binding and thereby serves well for protein screening. Specifically, SIRMS works remarkably like an ESI-MS, as it also needs a previous SEC and evaporation phase to transform the sample into a gaseous phase (Apostol et al., 2008). Nevertheless, the initial steps for SIRMS differ from the other methods: SIRMS labels proteins from different samples with heavy and light labels and consequently mixes them to use a tryptic solution to break down the samples. Lastly, the remaining steps are similar to the other procedures, where samples are exposed to the mentioned TOF-MS and classified based on their m/z ratio (von Neuhoff & Pich, 2005).

4.4 Surface plasmon resonance (SPR)-based detection methods

Surface plasmon resonance methods are yet another label-free detection method for the analysis of molecular interactions. They are usually considered optical techniques to monitor reactions and bindings, as their principles are based on light reflection, transfer of light energy to metal surfaces, and electromagnetism (Cooper, 2003; Nguyen et al., 2015). Moreover, they are accommodating in analyzing thermodynamic and kinetic interactions and thus serve as a perfect method for screening peptides in real-time (Syahir et al., 2015). Therefore, SPRs are considered biosensors and can be easily acquired by multiple instrumental companies all around the globe (Cooper, 2003; Nguyen et al., 2015; Syahir et al., 2015).

Currently, there are several variations of the base SPR method and its configuration but, all in all, the standard commercial setup of an SPR uses a geometry frequently called the *Kretschamann geometry* (Cooper, 2003; Nguyen et al., 2015). In this configuration, a gold surface gets hit by an incident light, which is condensed with the help of a prism or grating. Because of this, the energy of the light source is transferred from the photons of the condensed ray to the electrons of the metal surface, therefore making the electrons move and oscillate; however, this only happens at a certain incidence angle called the SPR angle and, as a result, produces a shadow in the reflected light (caused by the absorption of energy by the electrons) that is captured by a detector (Cooper, 2003; Nguyen et al., 2015). The mentioned electron movements result in the posterior generation of charged density waves, called *plasmons*, that propagate parallel to the gold surface and produce electromagnetic fields that are extremely sensible to changes in the dielectric constant of the medium (Cooper, 2003; Nguyen et al., 2015). If the light source is constant and the metal surface is thin, the SPR angle will depend completely on the reflective index of the medium close to the metal surface; therefore, if there is a small change in the reflective index of the medium (which commonly happens in molecular interactions as they affect the dielectric constant of the medium and thus, the reflective index), the plasmon stops forming at the initial angle and starts forming at a transposed angle. Detection of a binding is thus made by measuring the changes in the reflected light due to changes in the SPR angle caused by a chemical reaction (Cooper, 2003; Nguyen et al., 2015). The change in the refractive index can be also calculated with the following expression:

$$\Delta n_d = \left(\frac{dn}{dc}\right)_{vol} \frac{\Delta\Gamma}{h}$$

where $\left(\frac{dn}{dc}\right)_{vol}$ is the increase of refractive index n with the volume concentration of analyte c, $\Delta\Gamma$ is the concentration of the bound target on the surface, and Δn_d is the change in the refractive index (Nguyen et al., 2015).

Real-time detection is also possible with this technique. If a functional coating is placed on the metal surface receiving the incident light, various ligands or probes of interest can be immobilized in the coating with the help of molecular linkers (Cooper, 2003; Schasfoort, 2017). Afterward, a specific analyte can be studied for affinity to the immobilized ligand or probe if the constant flow of the molecule of interest passes close to the functional coating in the gold surface. Thus, a plot of the functional response vs. time can be done just by tracking the change in the refraction index and the SPR angle through time. The resulting graph will show a curve that accounts for the number of molecules bound to the probes and will give information about the reaction kinetics between ligand and analyte. Furthermore, the affinity of the analyte can be calculated with a ratio of the rate dissociation and association constants that can be obtained from the mentioned plot (Cooper, 2003; Nguyen et al., 2015; Schasfoort, 2017).

As can be seen, this versatile methodology opens the doors for several modifications to the original technique such as using microfluidic devices for the flow channels, using two-dimensional periodic arrays of different-sized gold nanoparticles as metal surfaces, or even modifying the whole geometry and data processing to create a high-throughput technique. As such, conventional SPR is not meant for high-throughput assays and multiplex analysis, as there are only maximum four flow cells attached to only one gold surface, which makes the technique slow for analyzing thousands of samples at a time (Cooper, 2003; Nguyen et al., 2015; Schasfoort, 2017). Nevertheless, a new SPR method called SPR Imaging (SPRI) emerged with its principal modification being the use of a charge-coupled device (CCD) camera that can vitalize a whole biochip. Therefore, biochips can be prepared in an array-like format where each array spot provides a lot of SPR data compared to the conventional SPR biosensing technique. Further modifications in the SPRI method include using a coherent polarized light beam to cover a larger surface area of analysis, using a constant SPR angle, and maintaining a constant light wavelength (Cooper, 2003; Nguyen et al., 2015; Schasfoort, 2017). It is clear, then, that this technique is of enormous value for peptide screening if a probe of interest (say, a cancer membrane protein) is immobilized in the functional coating and consequently exposed to a whole peptide library. This can allow the analysis of which peptides can effectively recognize and join to this probe and which peptides cannot (Cooper, 2003; Nguyen et al., 2015; Schasfoort, 2017).

5 Proven schemes for screening peptide libraries

5.1 High-throughput screening (HTS)

Screening aims to find chemicals in a library that exhibit a specific interaction with a chosen system (Entzeroth et al., 2009). In the context of peptide libraries screening describes the search for peptides that potentially have a desired activity on a specific microorganism or other type of biological system. HTS was developed in the early 1990s, and it is an experimental procedure in which thousands of small molecule compounds are simultaneously screened in a short period. When more than 100,000 are screened per day, it is referred to as ultra-HTS (uHTS) (Attene-Ramos et al., 2014).

In HTS for peptide libraries, the assay refers to an experiment that measures the effect of peptides on the system of interest (Wildey et al., 2017). It should be designed so that it ensures that peptides with the intended biological effect are identified during the screening process, and it needs to show a low level of variability and a high signal-to-background ratio, as this helps to avoid misleading results (Macarrón & Hertzberg, 2009). Conversely, having access to peptide libraries that are highly diverse and contain sufficient quantities of compounds of adequate quality is crucial for the successful identification of potential candidate peptides during HTS (Entzeroth et al., 2009).

For the process of HTS, first, the target is selected, which most of the time is a cell membrane receptor, but could also be an enzyme or DNA, for instance. After that, a large peptide library is assembled, and then the assay is designed and validated, considering that it must be aligned with the chosen target to create a robust and biologically relevant screen. Following, the peptides are screened in the assay. The screen must have enough throughput and low cost to allow the screening of large compound collections. Peptides that show antimicrobial potential are referred to as "hits." After identifying the hits in the initial screening, additional analysis can be conducted through HTS to gain insights into how to optimize their potential (Bokhari & Albukhari, 2022; Macarrón & Hertzberg, 2009).

Several types of assays can be developed depending on the chosen target; they can be divided into two big groups: biochemical assays and cell-based assays. Biochemical screening involves the use of a specific purified target protein, and it measures the interaction between the protein and other molecules, either through their binding or by inhibiting the protein's enzymatic activity in vitro. Typically, these assays are performed in a competitive manner where the compound being studied needs to replace a specific ligand or substrate. Usually, the result from the screening is read through an optical technique like fluorescence, absorbance, or luminescence. This kind of assay has a benefit in that all the successful findings are deliberately against a recognized target. Nonetheless, there might be a requirement for costly and laborious identification of the molecular mechanisms of action despite being aware of the target (Aldewachi et al., 2021; Blay et al., 2020).

Conversely, cell-based assays allow for the exploration of entire pathways, leading to the identification of several potential areas of interest instead of examining specific predetermined stages like in biochemical assays (Aldewachi et al., 2021). They include several classes of assays such as cell proliferation assays that allow for the observation of growth/no growth cellular responses, reporter gene assays that track cellular responses at translation or transcription level, protein fragment complementation assays, and two-hybrid screening with which it is possible to monitor the formation or inhibition of protein-protein interactions (Blay et al., 2020).

Screening of peptide libraries with HTS has been considered laborious, and with this technique, it is difficult to maintain scalability when trying to produce large quantities of peptides (da Cunha et al., 2017). Hence, there has been great interest in developing technologies for assay miniaturization, automation, and robotics. Miniaturization intends to diminish the number of samples needed. HT analysis is usually performed in 96- or 384-well microtiter plates (MTPs), and the culture is usually performed in deep 24-, 48-, and 96-well MTPs depending on the requirements of culture medium and oxygen (Zeng et al., 2020). Automation and robotics are important for improving scalability, especially when increasing the throughput of the method. Among the automated steps are sampling, fluorescence or color detection, data analysis, and collection of hits (Zeng et al., 2020). Robotic HTS systems can prepare, incubate, and analyze several MTPs simultaneously (Bogue, 2012).

5.2 Fluorescence-activated cell sorting (FACS)

FACS is a form of flow cytometry used to sort and count specific cells of interest from a diverse mixture of biological cells based on optical and fluorescent properties (Salam et al., 2023). This method was first created to extract mouse spleen cells attached to lactose-keyhole limpet hemocyanin by analyzing the amount of antigen produced by cells derived from antigen-binding cells (Carter et al., 2013). After the success of this, FACS has been used for sorting cell populations that can be readily distinguished from each other, such as hematopoietic or cultured cells (Carter et al., 2013). FACS as a screening method has enabled the evaluation and sorting of peptides based on their degree of expression, appropriate folding, and affinity for the target (Baeriswyl & Heinis, 2013).

The process of screening peptide libraries with FACS involves exposing the library to a soluble target that has been labeled with a fluorescent marker, and after this, cell sorting based on a flow cytometry system is performed (Bozovičar & Bratkovič, 2019). Flow cytometry functions by creating a suspension of individual cells from the sample being analyzed. For this, the sample of cells, previously stained with specific fluorescence, flows through a narrow stream of liquid and passes through a focused laser beam to measure the generated scattering light and excitation fluorescence of each cell. The strength of the fluorescence signals can be an indicator of the quantity of specific molecules within the cell nucleus or the degree to which certain antigens are expressed on the cell surface. After this measurement, a vibrating mechanism is applied to the sample, which separates the cells into droplets of individual cells. For cell sorting, an electric charge ring is used to create a charge based on the fluorescent measurement and an opposite charge, and as a droplet separates from the stream, it receives an opposing charge. These droplets are sorted and directed into separate containers based on their respective charges (Cai et al., 2020; Kaur et al., 2019). It is possible to perform cell sorting using a variety of collection systems, such as 96-well plates, different types of tubes, and microscopic slides, which have the appropriate growth medium in a solid or liquid state (Pereira et al., 2018).

FACS can be used for the screening of large libraries since it is a high-throughput technique exhibiting sorting rates of up to 100,000 cells/s and it can sort for different colored fluorophores separately, which allows it to track different parameters with a single assay. This screening method has the additional advantage of enabling the analysis of heterogeneous cell populations simultaneously, which is crucial for the optimal screening of peptide libraries across various types of cells (Kaur et al., 2019). Regardless, there is a major limitation associated with FACS, because it only allows measurement of intracellular or cellular-membrane-associated fluorescence, so the production of a secreted product will not be assessed, and this breaks the connection between genotype and phenotype that is required for FACS analysis (Bowman & Alper, 2020). Another drawback of FACS is the high cost of the equipment, but this limitation has been addressed using microfluidic platforms for FACS (μFACS) that besides lowering the cost of the process, enable parallelization (Bjork & Joensson, 2019).

5.3 Magnetic-activated cell sorting (MACS)

MACS is an affinity-based cell separation technique that, instead of using fluorescent labels like FACS, uses magnetic particles that can be modified to selectively bind to a specific subset of cells in a mixture (Shen et al., 2021). By binding only to the desired cells, MACS allows for the separation of the specific cell population from the rest of the mixture. This technique was introduced in the early 1990s by Miltenyi Biotec in Germany for the separation and analysis of various types of cells, and it remains the gold standard in magnetic cell separation because of its rapid and batch-wise processing (Plouffe et al., 2015; Wyatt Shields et al., 2015).

When screening peptide libraries with MACS, the library is incubated with magnetic beads coated with the target, and then there is a magnetic separation process that allows for the isolation of the cells of interest (Bozovičar & Bratkovič, 2019). Following the incubation of the cells with the magnetic-labeled peptides, the sample is passed through a magnetic field in a column. In this way, the labeled cells are trapped in the magnetic matrix since the magnetic field attracts the magnetized cells, and this allows them to be adsorbed onto the surface of the column, while the nonlabeled cells are washed away. After deactivating the magnetic field, the cells containing the beads are retrieved through elution. For magnetization of the peptides, colloidal superparamagnetic particles within the 20–100 nm range have become the preferred choice for magnetic cell separation due to their biological and optical inactivity (Fernandes et al., 2013; Rahmanian et al., 2017).

In MACS, the process of targeting cells and separating them are both based on magnetism principles, which enables the sorting process. As a result, MACS instruments available commercially are simpler, smaller, and significantly more affordable compared to FACS (Frenea-Robin & Marchalot, 2022). Furthermore, MACS allows for quicker processing of libraries compared to FACS, and multiple samples can be simultaneously processed in parallel (Salema & Fernández, 2017). This screening method has a throughput of approximately 10^9 cells per hour and it is highly specific; however, as FACS, it has the limitation that requires exogenous labeling (Plouffe et al., 2015). Lately, this technique has been combined with microfluidics, implementing it in a single microfluidic device, which offers greater precision in the separation of cells and allows for continuous flow processing by enhancing spatial and temporal control and it also improves the resolution of separation (Frenea-Robin & Marchalot, 2022).

5.4 Microarray screening

A microarray is a flat surface that displays molecules in a grid pattern, where each molecule is placed in a specific microscopic spot on the grid. This format enables fast parallel analysis of biomolecular interactions and reduces the amount of sample and analyte needed, since each microspot is an independent assay (Foong et al., 2012; Severgnini et al., 2011). This technique was developed in the early 1990s to quantify the transcription levels of multiple genes simultaneously, and since then, it has been widely used for the screening of several biomolecules such as DNA, RNA, and peptides (Berthuy et al., 2016).

Peptide microarrays can be synthesized either by in situ stepwise peptide synthesis or printed/spotted presynthesized peptides. The latter is also referred to as Merrifield solid-phase peptide synthesis (SPPS); in this method, peptides are assembled on a solid substrate, the amino groups of the side chains are removed, the peptides are cleaved from the substrate, and then they are immobilized to the array support (Szymczak et al., 2018). For the immobilization, peptides are deposited in small droplets using a robotic dispensing system, and there are several methods for the immobilization that are classified as noncovalent immobilization such as DNA/DNA and DNA/PNA, random covalent immobilization where the link is more stable than in noncovalent immobilization and it includes amine/epoxy and amine/NHS, and site-specific immobilization with which peptides are displayed in uniform orientations, examples of this linkage are native chemical ligation and Staudinger ligation (Sun et al., 2013). Substrate surface chemistry is critical in this method because it affects the effectiveness of peptide immobilization, surface density, and reaction activity, and it is important for the compatibility between the substrate and the peptides (Meng et al., 2018).

Conversely, in situ synthesis employs automated tools to perform simultaneous synthesis of peptides directly on the array (Szymczak et al., 2018). This method includes specialized techniques like SPOT-synthesis, which allows for the use of libraries of less than 10^3 peptides and is based on generating droplets of activated amino acids solutions and pouring them on a solid substrate like a glass slide or porous membrane; in this way, droplet adsorption creates a pattern of circular spots that can act as open reactors (López-Pérez et al., 2017; Wang & Distefano, 2014). Another method is photolithography, in which photogenerated reagents are used, most commonly photogenerated acids, that create an acidic environment when they are irradiated with light, which enables the formation of amide bonds (Katz et al., 2011). Particle-based synthesis uses a high-resolution printing technique, like laser printing, to deliver chemically activated amino acid monomers to a two-dimensional surface all at once, after printing the particles are melted, which releases the monomers and allows them to diffuse and join with the growing peptide chains on the substrate (Breitling et al., 2011).

After synthesis, the peptide microarray is incubated with the biological sample, and the molecular recognition events are detected either by label method or by label-free method (Brambilla et al., 2019). Label detection methods include fluorescent, electrochemical, chemiluminescent, and nanoparticle labeling, and label-free detection methods are based on molecular weight, refractive index, and molecular charge analysis (Syahir et al., 2015). These latter methods are considered more accurate since they reduce false positives when screening as the information is obtained directly from the characteristics of the peptides; however, label-based systems are still widely used because the equipment required is cost-effective, simpler, and easier to use compared to label-free systems (Chandra et al., 2011; Syahir et al., 2015).

6 Technical challenges in screening peptide libraries

6.1 Sensitivity and specificity

The screening of peptide libraries is a complex process that involves several technical challenges that can hinder the identification of bioactive peptides. One significant challenge is the limited sensitivity and specificity of screening methods, which can result in false positives and negatives (Wimley, 2019). To overcome this challenge, the screening conditions need to be optimized, considering factors such as pH, temperature, salt concentration, and reaction time, which can affect the binding affinity between the peptides and the target molecule (Mason et al., 2009). The quality of the library is also crucial, and it can be affected by factors such as peptide synthesis conditions, purification methods, and storage conditions; for example, degradation by proteases can result in false negatives (Getz et al., 2011), which can be prevented by using protease inhibitors and stabilization agents. Incorporating nonnatural amino acids or modifications can enhance the stability and bioactivity of the peptides. Additionally, the library's complexity and diversity can lead to a high level of background noise and false positives, which can be overcome by using miniaturized systems and advanced detection technologies (Battersby & Trau, 2002). The diversity of the library can also affect the identification of bioactive peptides, and iterative screening and optimization of the peptide structure can improve bioactivity and selectivity (Bozovičar & Bratkovič, 2019).

The sensitivity and specificity of peptide library screening methods can depend on the specific method used, and each method has its own set of advantages and limitations. Phage display and ribosome display both have limitations in terms of the size and complexity of the peptides that can be displayed, which can affect the diversity and quality of the library. Cell surface display methods require efficient surface expression of peptides, which can be affected by various factors such as folding, stability, and toxicity, while cDNA/mRNA display presents the challenge in the maintenance of protein activity during the screening process. Microfluidic devices and nanotechnology-based platforms also require optimization of various parameters to achieve optimal screening performance. The development of novel technologies and the optimization of existing methods are required to overcome these challenges and improve the screening performance of peptide libraries.

6.2 Integration of technologies

Hybrid approaches that combine different screening methods and technologies have been developed to overcome these limitations. However, careful optimization and standardization are required to ensure compatibility and reproducibility. Integration of technologies is a critical aspect of screening peptide libraries, as it requires the compatibility of various technologies used in the process; it can also facilitate the development of miniaturized and HTS assays, which can significantly improve the screening process efficiency and reduce costs. Standardization of protocols and techniques is necessary to ensure consistent and reproducible results across different laboratories and platforms. Finally, the collaboration between researchers and companies working on different aspects of the screening process can help to overcome technical challenges and accelerate the development of screening peptide libraries.

7 Future perspectives

Peptide library research has grown in popularity in recent years because it provides a powerful tool for identifying new peptides with a wide range of potential uses, including drug discovery, protein engineering, and biomaterials development. Screening is a major strategy in peptide library research that involves examining many peptides for specified properties or activities. The ramifications of screening in peptide library research are substantial in the future. Advances in HTS methods will allow researchers to screen vast libraries of peptides quickly and efficiently, enabling the identification of novel peptides with unparalleled speed and precision. The increasing availability of computational approaches for predicting peptide characteristics and functions will improve screening by allowing researchers to filter and prioritize peptide candidates for further study. This reduces the amount of time and resources needed for experimental validation, resulting in more efficient and cost-effective peptide discovery. New screening approaches that combine diverse and complicated biological systems, such as cell-based assays and in vivo models, will broaden the scope of peptide library research and allow for the identification of peptides with more complex functions and applications. Furthermore, the combination of screening techniques with other technologies, such as gene editing and synthetic biology, will enable the synthesis of custom-designed peptides with specified properties and functionalities, bringing peptide engineering to a new level of precision and control. Screening in peptide library research has the potential to transform the discipline and offer new paths for the creation of innovative peptide-based treatments and biomaterials.

8 AI disclosure

In the preparation of this chapter, artificial intelligence technology, specifically OpenAI's ChatGPT, was employed to assist with grammar corrections and the refinement of English language use. This technology provided suggestions on language structure, grammar, and style to enhance the readability and clarity of the content. Following the use of this tool, the content was thoroughly reviewed and edited as necessary. The author(s) take(s) full responsibility for the final content of this publication.

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ANTIMICROBIAL PEPTIDES

A Roadmap for Accelerating Discovery and Development

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Contains a comprehensive description of computational and experimental approaches for the discovery, manufacture, and clinical translation of antimicrobial peptides.

Antimicrobial Peptides: A Roadmap for Accelerating Discovery and Development covers the most important efforts of scientists and engineers worldwide to accelerate the process of discovery, production, and eventual market penetration of more potent antimicrobial peptides. These efforts have been fueled by emerging technologies such as artificial intelligence and data science, molecular and CFD simulations, easy-to-use process simulation packages, microfluidics, and 3D printing, among many others. Such technologies can now be implemented and scaled up quickly and at relatively low cost in low-budget production facilities, critical to moving to sustainable and marketable products worldwide.

Discovering novel antimicrobial peptides rationally and cost-effectively has emerged as one of the significant challenges of modern biotechnology. Thus far, this process has been tedious and costly, resulting in molecules with activities far below those needed to address the current challenge of microbial resistance to antibiotics that takes the lives of thousands of people around the world every year. Due to these issues, large biopharmaceutical companies have decided not to invest in discovering novel antimicrobials, which is worrisome in the long run due to the possibility of a new pandemic.

This book also delineates how multidisciplinary teams have assembled to address the challenges of manufacturing, biological testing, and clinical trials to finally reach complete translation. It is an ideal resource for scientists, researchers, engineers, and students in the fields of biotechnology, microbiology, biochemistry, and pharmaceuticals. Finally, anyone interested in the emerging technologies that are enabling the discovery and development of new antimicrobial peptides may find this book interesting and informative.

Key Features

- Covers computational tools (including emerging artificial intelligence algorithms) and microfluidic systems for discovering and high-throughput screening of AMPs
- Discusses the application of bioprocess engineering scale-up approaches for AMPs production and purification with the aid of process simulation tools and rapid prototyping
- Highlights user-centered design and formulation of products with AMPs
- Describes the whole pipeline for AMPs production

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